

Explicit dissociated sets from tensor lattices: primitivity certificates and an effective decay rate

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Abstract

A finite set of positive integers is *dissociated* if its subset sums are pairwise distinct. We give explicit constructions from the tensor lattices underlying the recent disproof of Erdős’s distinct subset sums conjecture by GPT-6 Astra, as expounded by Bloom. A prescribed perturbation of a lattice in dimension $d = b^s$ has a normal vector factoring into s vectors of length b . We prove an exact formula for its saturation index in terms of the contents (the gcds of the coefficients) of s circulant adjugates, and characterise primitivity by polynomial gcds over prime fields. Thus large constructions can be certified through small matrices.

Exact integer computations give a dissociated set of size $n = 43\,923$ with $\max A < 0.158523 \cdot 2^n$, and one of size $n = 3\,841\,966$ with $\max A < 0.103037 \cdot 2^n$, improving Bohman’s explicit constant 0.22002. Each bound extends to every larger size by doubling and adjoining 1.

For every odd $b \geq 5$ with $3 \nmid b$, we prove that the relevant first-level resultant is nonzero. The proof reduces a hypothetical common root to a nonzero algebraic integer all of whose conjugates have absolute value less than 1. This nonvanishing controls every tensor depth: the scalings $Q = 2^{s+1}t$ are primitive whenever t is a sufficiently large multiple of a modulus depending only on b . Explicit bounds on this modulus give the unconditional rate $f(n) \leq n^{-1/(50 \log \log n)}$ for all sufficiently large n , where $f(n)$ is the infimum of $\max A/2^{n-1}$ over dissociated sets of size n .

Mathematics Subject Classifications: 11B75, 11H06, 11H31, 11Y16

1 Introduction

1.1 Background

Let $A \subset \mathbb{N}$ be finite. We call A *dissociated* (or say that A has *distinct subset sums*) if the map $S \mapsto \sum_{a \in S} a$ is injective on the subsets $S \subseteq A$. Erdős, who called this “perhaps my first serious problem”, conjectured that every dissociated $A \subseteq \{1, \dots, N\}$ with $|A| = n$

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satisfies $N \gg 2^n$, and offered \$500 for a proof or disproof; it is Problem 1 of the collection [3]. Writing

$$f(n) := \inf \left\{ \frac{\max A}{2^{n-1}} : A \subset \mathbb{N} \text{ dissociated, } |A| = n \right\},$$

the powers of two give $f(n) \leq 1$. Until 2026, every improvement of this trivial bound came from an explicit construction, and each such bound propagates from one size to all larger sizes because f is nonincreasing (Lemma 23 below). The Conway–Guy sequence [7, 11], shown to consist of dissociated sets by Bohman [4], gives $N < 0.23513 \cdot 2^n$ for $n \geq 40$; Lunnon’s computer search [13] gives $N < 0.22096 \cdot 2^n$ for $n \geq 67$; and Bohman [5] obtained $f(n) \leq 0.44004$ for large n , i.e. $N \leq 0.22002 \cdot 2^n$, the best explicit construction prior to the present paper. In the other direction Erdős and Moser [10] proved $N \geq (\frac{1}{4} - o(1))2^n / \sqrt{n}$; the constant was improved several times (Elkies [9], Aliev [2]; see the table in [14] for the full history), the current record $\sqrt{2/\pi}$ being due to Elkies and Gleason (unpublished) and to Dubroff, Fox and Xu [8], who prove $N \geq \binom{n}{\lfloor n/2 \rfloor}$.

In September 2026 the conjecture was *disproved*: for every $\varepsilon > 0$ there are arbitrarily large n with $f(n) < \varepsilon$. The proof was found by GPT-6 Astra and verified in Lean [1]. We adopt the lattice notation of Bloom’s exposition [3]. Its first step constructs lattices $\Lambda_s \subset V_d = \{x \in \mathbb{R}^d : \sum_i x_i = 0\}$, $d = b^s$, which avoid the open unit cube and have small covolume. Its second step approximates $Q\Lambda_s$ by a primitive lattice $L = u^\perp \cap \mathbb{Z}^d$ and transfers this to dissociated integers. Bloom invokes primitive-lattice equidistribution for that step. Horesh and Karasik [12] give quantitative error terms, but the exposition does not extract the dimension-dependent constants needed for a rate.

The original formal proof already contains a different algebraic passage to primitive lattices. The module cited in [1] uses integral Smith-type diagonalisation, a saturated bidiagonal perturbation and explicit normal weights; its perturbation buffer is obtained for each fixed matrix. It does not state a quantitative bound on n in terms of ε . Our contribution is a perturbation that retains the tensor coordinates: its normal vector and saturation index reduce to s calculations in dimension b , with an explicit perturbation constant and local tests that yield the numerical constructions below. A new nonvanishing theorem for the first-level resultant then provides arithmetic progressions of primitive scalings at every depth, with a modulus whose size we control explicitly. This yields an unconditional effective decay rate for the tensor construction. The classical monotonicity lemma below already combines with the original disproof to give qualitative convergence $f(n) \rightarrow 0$ along all n .

1.2 Results

We develop and certify the tensor-preserving perturbation just described. Throughout, $b \geq 5$ is odd with $3 \nmid b$, $s \geq 1$, $d = b^s$, $R_s = \mathbb{Z}[x_1, \dots, x_s] / (x_1^b - 1, \dots, x_s^b - 1) \cong \mathbb{Z}^d$, $g(x) = (2 + x)(1 - x) = 2 - x - x^2$, and

$$\Delta_{b,s} := \frac{(1 + 2^{-b})^{(b^s - 1)/(b - 1)}}{(3/2)^s} \tag{1}$$

is the normalised covolume $\text{covol}(\Lambda_s) / \sqrt{d}$ of the construction (Proposition 9).

Closed form of the lattices. We show (Proposition 9) that

$$\Lambda_s = \bigoplus_{k=1}^s 2^{-k} (2 + x_1) \cdots (2 + x_k) (1 - x_k) \mathbb{Z}[x_k, \dots, x_s], \quad v_s = 2^{-s} \prod_{k=1}^s (2 + x_k),$$

where $\mathbb{Z}[x_k, \dots, x_s] \subset R_s$ is the span of the monomials not involving $x_1, \dots, x_{k-1}$.

An explicit primitivisation. For $Q \in 2^s \mathbb{Z}$ we add to each basis vector $Q 2^{-k} (2 + x_1) \cdots (2 + x_k) (1 - x_k) x_k^i m$ of $Q \Lambda_s$ ($0 \leq i \leq b-2$, m a monomial in $x_{k+1}, \dots, x_s$) the monomial $x_1^{b-1} \cdots x_{k-1}^{b-1} x_k^i m$. This defines a lattice $L_s(Q) \subset \mathbb{Z}^d$ of rank $d-1$. Define *level polynomials* $\varphi_k = \theta_k g + \rho_k \in \mathbb{Z}[x]/(x^b - 1)$ recursively by $\theta_1 = Q/2$, $\rho_1 = 1$, $\psi_k = \text{adj}(\overline{\varphi_k})$ (the adjugate element, $\psi_k \overline{\varphi_k} = N(\varphi_k)$), $\gamma_k = \text{cont}(\psi_k)$, $\tilde{u}_k = \pm x^{b-1} \psi_k / \gamma_k$, $a_k = \langle \tilde{u}_k, x^{b-1} \rangle$, $c_k = \langle \tilde{u}_k, 2 + x \rangle$, $\theta_{k+1} = 2^{-(k+1)} Q c_1 \cdots c_k$, $\rho_{k+1} = a_1 \cdots a_k$ (Definition 14). The constant K_s below is explicit (Proposition 12).

Theorem 1 (normal vector and index). *Let $Q \in 2^s \mathbb{Z}$ with $Q \geq 8K_s$. Then $L_s(Q)$ has rank $d-1$, its orthogonal complement is spanned by the primitive integer vector $u = \tilde{u}_1 \otimes \cdots \otimes \tilde{u}_s$, i.e. $u(x) = \prod_k \tilde{u}_k(x_k)$, all coordinates of u are positive, and*

$$[\text{sat } L_s(Q) : L_s(Q)] = \prod_{k=1}^s \gamma_k^{b^{s-k}}.$$

In particular $L_s(Q)$ is primitive if and only if $\gamma_k = \text{cont } \text{adj}(\overline{\varphi_k}) = 1$ for every k .

Theorem 2 (level criterion and exceptional primes). *For $\varphi = \theta g + \rho \in \mathbb{Z}[x]/(x^b - 1)$ with $N(\varphi) \neq 0$, one has $\text{cont } \text{adj}(\overline{\varphi}) = 1$ if and only if $\deg \gcd_{\mathbb{F}_p}(\varphi, x^b - 1) \leq 1$ for every prime p . Let P_b be the set of odd primes dividing $b(2^b + 1)n_b$, where $n_b = \prod_{0 < i < j < b} \text{Res}_z\left(\frac{z^b - 1}{z - 1}, 1 + z^i + z^j\right) \neq 0$. If p is an odd prime with $p \notin P_b$ and $p \nmid \gcd(\theta, \rho)$, then $\deg \gcd_{\mathbb{F}_p}(\varphi, x^b - 1) \leq 1$.*

Theorem 3 (first level, unconditional). *Let $q \geq 4K_1$ be an integer with $\text{rad}(P_b) \mid q$. Then $L_1(2q)$ is primitive; the coordinates of $\tilde{u}_1$ are positive integers forming a $(2q - K_1)$ -dissociated set, with $\max \tilde{u}_1 \leq (2q)^{b-1} \Delta_{b,1} (1 + 5K_1/q)$. Consequently, for every l with $2^l \leq 2q - K_1$ the set $\{2^j \tilde{u}_{1,i} : 0 \leq j < l, 0 \leq i < b\}$ is dissociated of size lb , and $f(lb) \leq \max \tilde{u}_1 / 2^{l(b-1)}$.*

Explicit dissociated sets. Theorem 1, a perturbation lemma (Lemma 10) and a size estimate (Lemma 11) together give the following.

Theorem 4. *Let $2^s \mid Q$, $Q \geq 8K_s$, and suppose $\gamma_k = 1$ for $k = 1, \dots, s$ (a finite computation). Then the $d = b^s$ coordinates of u are positive integers forming a $(Q - K_s)$ -dissociated set, and*

$$\max u = \prod_{k=1}^s \max \tilde{u}_k \leq Q^{d-1} \Delta_{b,s} (1 + 10K_s/Q).$$

For every l with $2^l \leq Q - K_s$ the set $A = \{2^j u_i\}$ is dissociated, $|A| = ld$, and $f(ld) \leq \max u / 2^{l(d-1)}$.

The certificate $\gamma_1 = \cdots = \gamma_s = 1$ requires s adjugate computations of $b \times b$ integer matrices; the integers grow with d , but a full $d \times d$ computation is avoided. Table 1 lists the outcomes; for example

$$\begin{aligned} (b, s) = (11, 3) : \quad n &= 43\,923, & \max A &< 0.158523 \cdot 2^n, \\ (b, s) = (13, 4) : \quad n &= 1\,199\,562, & \max A &< 0.132535 \cdot 2^n, \\ (b, s) = (17, 4) : \quad n &= 3\,841\,966, & \max A &< 0.103037 \cdot 2^n. \end{aligned}$$

These explicit dissociated sets improve Bohman's constant 0.22002. By Lemma 23 each bound persists for all larger sizes: there are explicit dissociated sets with $\max A < 0.158523 \cdot 2^n$ for every $n \geq 43\,923$ and with $\max A < 0.103037 \cdot 2^n$ for every $n \geq 3\,841\,966$ (Corollary 24).

Good scalings at every tensor depth. Write $Q = 2^{s+1}t$. Before contents are normalised, the level data $\hat{a}_k, \hat{c}_k$ are polynomials in t with integer coefficients (Section 5.1). Let $H(b, s)$ be the statement that $\hat{a}_k$ and $\hat{c}_k$ are coprime in $\mathbb{Q}[t]$ for $1 \leq k \leq s-1$.

Theorem 5. *For every allowed b there is an integer $M_b \geq 1$, independent of s , such that $L_s(2^{s+1}t)$ is primitive for every $t \equiv 0 \pmod{M_b}$ with $2^{s+1}t \geq 8K_s$. Consequently $\limsup_{l \rightarrow \infty} f(lb^s) \leq \Delta_{b,s}$.*

The key step is Theorem 26: the first-level resultant r_b of (7) is nonzero for every allowed b . We then prove $H(b, s)$ at every depth by a homogeneous recursion (Proposition 27), and may take $M_b = \text{rad}(P_b)|r_b|$. Thus, for example, $\limsup_l f(1331l) \leq \Delta_{11,3} < 0.31618$ and $\limsup_l f(2197l) \leq \Delta_{13,3} < 0.30299$ hold unconditionally. The explicit estimate $\log M_b \leq 4b^3$ gives $f(n) \leq n^{-1/(20 \log \log n)}$ for infinitely many n (Theorem 33), and $f(n) \leq n^{-1/(50 \log \log n)}$ for all sufficiently large n (Corollary 34). Both statements are proved in full in Section 5.1.

Sizes and heights. The classical doubling step (Lemma 23) fills gaps between the sizes $n = ld$ without loss in the normalised maximum. Section 7 records the exact costs of other padding and lifting operations. We also relate decay to the height of admissible pairs: a height $h \geq b^\alpha$ and sufficiently small excess covolume give a tensor covolume $O(d^{-\alpha})$. Transferring this to $f(n) \leq n^{-\alpha+o(1)}$ requires quantitative control of primitivisation and of the available sizes, which we state explicitly there. Vaaler's theorem bounds the height by $O(\sqrt{b})$ when the full lattice covolume is bounded (Proposition 35).

Organisation. Section 2 recasts the lattices of [3] in the polynomial model R_s , in which a cyclic shift of coordinates is multiplication by a variable, and proves the closed form. Section 3 contains the two perturbation lemmas and the constant K_s . Section 4 defines $L_s(Q)$ and proves Theorems 1, 2 and 3. Section 5 deduces Theorems 4 and 5, the monotonicity lemma with its corollaries, and the effective decay rate. Section 6 describes the computations behind Table 1, and Section 7 discusses the sizes reached by the construction, the height problem and Siegel's lemma.

1.3 Notation and conventions

For a lattice L of rank r in $\mathbb{R}^d$, $\text{covol}(L)$ is the r -dimensional volume of a fundamental domain. A sublattice $L \subseteq \mathbb{Z}^d$ is *primitive* (or saturated) if $L = (L \otimes \mathbb{R}) \cap \mathbb{Z}^d$; we write $\text{sat } L$ for $(L \otimes \mathbb{R}) \cap \mathbb{Z}^d$. If $L \subset \mathbb{Z}^d$ has rank $d-1$ with basis matrix $B \in \mathbb{Z}^{d \times (d-1)}$, the vector w of signed maximal minors of B spans $L^\perp$ and $\text{gcd}(w) = [\text{sat } L : L]$; a primitive lattice of rank $d-1$ is $R(u) := u^\perp \cap \mathbb{Z}^d$ for a primitive $u \in \mathbb{Z}^d$, and $\text{covol}(R(u)) = |u|_2$. For $u \in \mathbb{Z}^d$ and real $Q > 0$, the coordinates of u are Q -*dissociated* if the only $c \in \mathbb{Z}^d$ with $|c_i| < Q$ for all i and $\sum_i c_i u_i = 0$ is $c = 0$; equivalently $R(u) \cap (-Q, Q)^d = \{0\}$. The *lift* of a 2^l -dissociated vector u with positive coordinates is $A = \{2^j u_i : 0 \leq j < l, 1 \leq i \leq d\}$; two subset sums of A are $\sum_i c_i u_i$ and $\sum_i c'_i u_i$ with $0 \leq c_i, c'_i < 2^l$, so A is dissociated, $|A| = ld$, $\max A = 2^{l-1} \max u$, and

$$f(ld) \leq \frac{2^{l-1} \max u}{2^{ld-1}} = \frac{\max u}{2^{l(d-1)}}. \quad (2)$$

$R_b = \mathbb{Z}[x]/(x^b - 1)$ is identified with $\mathbb{Z}^b$ via $x^i \leftrightarrow e_i$; the standard inner product becomes $\langle f, g \rangle = [x^0](f\bar{g})$, where $\bar{g}(x) = g(x^{-1})$ and $[x^i]h$ denotes the coefficient of x^i in h ; in particular $\langle f, x^i \rangle = [x^i]f$. For $a \in R_b$ let m_a be the $b \times b$ matrix of multiplication by a (a circulant), $N(a) = \det m_a = \prod_{\zeta^b=1} a(\zeta)$, and $\text{adj}(a) \in R_b$ the element with $m_{\text{adj}(a)} = \text{adj}(m_a)$, so that $a \text{adj}(a) = N(a)$; since $\text{adj}(m_a)$ commutes with every m_c , it is indeed a multiplication matrix. Note $m_a^T = m_{\bar{a}}$. The content $\text{cont}(\psi)$ of $\psi \in R_b$ is the gcd of its coefficients. Finally $\mathbf{1}_d = (1, \dots, 1) \in \mathbb{R}^d$, $V_d = \mathbf{1}_d^\perp$, and μ_b is the group of complex b -th roots of unity.

2 The lattices

2.1 Admissible pairs

Following [3], a pair (Λ, v) with $\Lambda \subset V_d$ a lattice of rank $d-1$ and $v \in \mathbb{R}^d \setminus V_d$ is *admissible* if

$$(A1) \quad \Lambda \cap (-1, 1)^d = \{0\};$$

$$(A2) \quad tv \in \Lambda + (-1, 1)^d \text{ with } t \in \mathbb{R} \text{ implies } |t| < 1.$$

Both conditions are invariant under $v \mapsto -v$; replacing v by $-v$ if necessary, we always orient an admissible pair so that its *height* $h = \langle v, \mathbf{1}_d \rangle$ is positive. Then $\Gamma = \Lambda \oplus \mathbb{Z}v$ satisfies $\text{covol}(\Gamma) = \text{covol}(\Lambda) h / \sqrt{d}$. We write $\Delta = \text{covol}(\Lambda) / \sqrt{d} = \text{covol}(\Gamma) / h$. Conditions (A1)–(A2) imply $\Gamma \cap (-1, 1)^d = \{0\}$, hence $\text{covol}(\Gamma) \geq 1$ by Minkowski's theorem, so $\Delta \geq 1/h$.

The base pair. Let $b \geq 3$ be odd. Following [3], put

$$\Lambda_1 = \tfrac{1}{2}gR_b = \tfrac{1}{2}(2+x)((1-x)R_b), \quad v_1 = \tfrac{1}{2}(2+x).$$

Identify R_b with $\mathbb{Z}^b$, index coordinates by $0, \dots, b-1$, and let $M = I + \tfrac{1}{2}P$, where $(Py)_i = y_{i-1}$ with subscripts taken modulo b . Then $(1-x)R_b = \mathbb{Z}^b \cap V_b$, so $\Lambda_1 = M(\mathbb{Z}^b \cap V_b)$ and $v_1 = Me_0$.

Lemma 6. *The pair (Λ_1, v_1) is admissible. Its invariants are*

$$h_1 = \frac{3}{2}, \quad \text{covol}(\Gamma_1) = 1 + 2^{-b}, \quad \Delta_{b,1} = \frac{2}{3}(1 + 2^{-b}).$$

Proof. For every $y \in \mathbb{R}^b$,

$$\|My\|_\infty \geq \|y\|_\infty - \frac{1}{2}\|Py\|_\infty = \frac{1}{2}\|y\|_\infty.$$

Thus $My \in (-1, 1)^b$ implies $|y_i| < 2$ for every i .

For (A1), suppose $z \in \mathbb{Z}^b \cap V_b$ and $Mz \in (-1, 1)^b$. Each z_i belongs to $\{-1, 0, 1\}$. If $z_i = 1$, the inequality $|z_i + z_{i-1}/2| < 1$ forces $z_{i-1} = -1$; if $z_i = -1$, it forces $z_{i-1} = 1$. Hence a nonzero coordinate would force alternating signs around the entire cycle, which is impossible because b is odd. Therefore $z = 0$.

For (A2), suppose $t \in \mathbb{R}$, $z \in \mathbb{Z}^b \cap V_b$, and $tv_1 - Mz \in (-1, 1)^b$. Set $y = te_0 - z$. Then $My \in (-1, 1)^b$, the coordinates $y_1, \dots, y_{b-1}$ belong to $\{-1, 0, 1\}$, and $\sum_i y_i = t$. We first show $|y_0| < 1$. If $y_0 \geq 1$, the coordinate-0 inequality forces $y_{b-1} = -1$. Applying the same sign rule successively backwards forces $y_1 = 1$, since $b-1$ is even. But then $y_1 + y_0/2 \geq 3/2$, contradicting the coordinate-1 inequality. Replacing y by $-y$ rules out $y_0 \leq -1$.

If one of $y_1, \dots, y_{b-1}$ is zero, every following coordinate in this list is zero, since an integer y_{i+1} satisfying $|y_{i+1} + 0/2| < 1$ must vanish. The nonzero coordinates in the list therefore form an initial alternating sequence. If this sequence is nonempty, its first sign is opposite to that of y_0 , by $|y_1 + y_0/2| < 1$. Its sum is either 0 or y_1 . In the former case $|t| = |y_0| < 1$; in the latter case $t = y_0 + y_1$ also lies in $(-1, 1)$. This includes the empty sequence and proves (A2) for every real t .

Finally, $h_1 = \langle Me_0, \mathbf{1}_b \rangle = 3/2$, and $(\mathbb{Z}^b \cap V_b) \oplus \mathbb{Z}e_0 = \mathbb{Z}^b$ gives $\Gamma_1 = M\mathbb{Z}^b$. Since b is odd,

$$\det M = 1 - (-1/2)^b = 1 + 2^{-b}.$$

The formula for $\Delta_{b,1}$ follows by dividing by h_1 . □

2.2 Tensor products of admissible pairs

Proposition 7. *Let (Λ, v) be admissible in $\mathbb{R}^b$ and (Λ', v') admissible in $\mathbb{R}^{d'}$. In $\mathbb{R}^b \otimes \mathbb{R}^{d'} = \mathbb{R}^{bd'}$ put*

$$\Lambda'' = \Lambda \otimes \mathbb{Z}^{d'} \oplus v \otimes \Lambda', \quad v'' = v \otimes v'.$$

Then (Λ'', v'') is admissible, of height hh' , and $\text{covol}(\Gamma'') = \text{covol}(\Gamma)^{d'} \text{covol}(\Gamma')$.

Proof. View elements of $\mathbb{R}^b \otimes \mathbb{R}^{d'}$ as $b \times d'$ matrices. An element of Λ'' has the form $X = \sum_i \beta_i z_i^T + v \lambda^T$ with $\beta_i \in \Lambda$, $z_i \in \mathbb{Z}^{d'}$, $\lambda \in \Lambda'$, so its m -th column is $\sum_i \beta_i z_{i,m} + \lambda_m v \in \Lambda + \lambda_m v$. Coordinate sums vanish, and $\Lambda \otimes \mathbb{R}^{d'}$, $v \otimes \mathbb{R}^{d'}$ intersect trivially, so Λ'' is a lattice of rank $(b-1)d' + (d'-1) = bd' - 1$ in $V_{bd'}$. If $X \in (-1, 1)^{bd'}$, each column lies in $(\Lambda + \lambda_m v) \cap (-1, 1)^b$, so $|\lambda_m| < 1$ by (A2) for (Λ, v) ; thus $\lambda \in \Lambda' \cap (-1, 1)^{d'} = \{0\}$, and then every column lies in $\Lambda \cap (-1, 1)^b = \{0\}$. If $X \in \Lambda'' + tv''$ has all entries in $(-1, 1)$, then every column lies in $\Lambda \cap (-1, 1)^b = \{0\}$.

the m -th column lies in $\Lambda + (\lambda_m + tv'_m)v$, so $|\lambda_m + tv'_m| < 1$ for all m , i.e. $\lambda + tv' \in (\Lambda' + tv') \cap (-1, 1)^{d'}$, and (A2) for (Λ', v') gives $|t| < 1$. The height is $\langle v, \mathbf{1}_b \rangle \langle v', \mathbf{1}_{d'} \rangle = hh'$. Finally $\Gamma'' = \Lambda \otimes \mathbb{Z}^{d'} \oplus v \otimes \Gamma'$; the first summand spans $V_b \otimes \mathbb{R}^{d'}$ and has covolume $\text{covol}(\Lambda)^{d'}$, while the projection of $v \otimes \Gamma'$ onto $(V_b \otimes \mathbb{R}^{d'})^\perp = \mathbf{1}_b \otimes \mathbb{R}^{d'}$ is $\frac{h}{\sqrt{b}} \hat{n} \otimes \Gamma'$ with $\hat{n} = \mathbf{1}_b / \sqrt{b}$, of covolume $(h/\sqrt{b})^{d'} \text{covol}(\Gamma')$. Hence $\text{covol}(\Gamma'') = (\text{covol}(\Lambda)h/\sqrt{b})^{d'} \text{covol}(\Gamma') = \text{covol}(\Gamma)^{d'} \text{covol}(\Gamma')$. $\square$

The iteration of [3] is the same operation with the roles of the two factors exchanged; both produce admissible pairs with identical invariants. We use the order of Proposition 7 because it gives a clean closed form.

Definition 8. (Λ_1, v_1) is the base pair in the variable x_1 ; for $s \geq 2$, $(\Lambda_s, v_s) := (\Lambda_1, v_1) \otimes (\Lambda_{s-1}, v_{s-1})$, where Λ_{s-1} lives in the variables $x_2, \dots, x_s$.

Proposition 9 (closed form). *Let $\mathbb{Z}[x_k, \dots, x_s] \subset R_s$ be the span of the monomials in $x_k, \dots, x_s$. Then*

$$\Lambda_s = \bigoplus_{k=1}^s \Lambda_s^{(k)}, \quad \Lambda_s^{(k)} = 2^{-k} (2 + x_1) \cdots (2 + x_k) (1 - x_k) \mathbb{Z}[x_k, \dots, x_s],$$

$v_s = 2^{-s} \prod_{k=1}^s (2 + x_k)$, $h_s = (3/2)^s$, $\text{covol}(\Gamma_s) = (1 + 2^{-b})^{(b^s - 1)/(b - 1)}$ and $\text{covol}(\Lambda_s)/\sqrt{b^s} = \Delta_{b,s}$ as in (1). A basis of $\Lambda_s^{(k)}$ consists of the vectors $2^{-k} (2 + x_1) \cdots (2 + x_k) (1 - x_k) x_k^i m$, $0 \leq i \leq b - 2$, m a monomial in $x_{k+1}, \dots, x_s$; the denominators of Λ_s divide 2^s .

Proof. Induction on s ; $s = 1$ is the definition of Λ_1 , since $(1 - x)R_b$ has basis $(1 - x)x^i$, $i \leq b - 2$. For $s \geq 2$, $\Lambda_1 \otimes \mathbb{Z}[x_2, \dots, x_s] = \frac{1}{2}(2 + x_1)(1 - x_1)R_s = \Lambda_s^{(1)}$ and $v_1 \otimes \Lambda_{s-1} = \frac{1}{2}(2 + x_1) \bigoplus_{k \geq 2} 2^{-(k-1)} (2 + x_2) \cdots (2 + x_k) (1 - x_k) \mathbb{Z}[x_k, \dots, x_s] = \bigoplus_{k \geq 2} \Lambda_s^{(k)}$ by the induction hypothesis. The invariants follow from Proposition 7: $h_s = \frac{3}{2}h_{s-1}$ and $\text{covol}(\Gamma_s) = \text{covol}(\Gamma_1)^{b^{s-1}} \text{covol}(\Gamma_{s-1})$, which gives the exponent $1 + b + \cdots + b^{s-1}$. $\square$

3 Perturbation lemmas

Let (Λ, v) be admissible in $\mathbb{R}^d$, let $b_1, \dots, b_{d-1}$ be a fixed basis of Λ , and let $b_1^*, \dots, b_{d-1}^* \in V_d$ be the dual basis, so that $\langle b_i, b_j^* \rangle = \delta_{ij}$. Put

$$K := \sum_{j=1}^{d-1} |b_j^*|_1. \quad (3)$$

For $\lambda = \sum_j m_j b_j \in \Lambda$ we have $m_j = \langle \lambda, b_j^* \rangle$, hence

$$\sum_j |m_j| \leq K |\lambda|_\infty. \quad (4)$$

Let $Q > 0$ be real, let $r_1, \dots, r_{d-1} \in \{1, \dots, d\}$ be *distinct* rows, and

$$L := \text{span}_{\mathbb{Z}} \{ \ell_j := Qb_j + e_{r_j} : 1 \leq j \leq d - 1 \} \subset \mathbb{R}^d.$$

Lemma 10 (cube avoidance). *If $Q > K$ then the ℓ_j are linearly independent and $L \cap (-Q - K, Q - K)^d = \{0\}$. In particular, if $L \subseteq \mathbb{Z}^d$ is primitive with normal vector u , the coordinates of u are $(Q - K)$ -dissociated.*

Proof. Let $m \in \mathbb{R}^{d-1}$, $x = \sum_j m_j \ell_j = Q\lambda + \sum_j m_j e_{r_j}$ with $\lambda = \sum_j m_j b_j$. Since $m_j = \langle \lambda, b_j^* \rangle$, the bound (4) holds for real m as well, so if $m \neq 0$ then $\lambda \neq 0$ (the b_j are a basis) and $|x|_\infty \geq Q|\lambda|_\infty - \sum_j |m_j| \geq (Q - K)|\lambda|_\infty > 0$; hence the ℓ_j are linearly independent over $\mathbb{R}$. If moreover $m \in \mathbb{Z}^{d-1} \setminus \{0\}$, then $\lambda \in \Lambda \setminus \{0\}$, so $|\lambda|_\infty \geq 1$ by (A1) and $|x|_\infty \geq Q - K$. $\square$

Lemma 11 (shape of the normal vector). (i) *Let $Q > K$ and let $u \in \mathbb{R}^d \setminus \{0\}$ be orthogonal to L . Then $\langle u, \mathbf{1}_d \rangle \neq 0$ and, after replacing u by $-u$ if necessary, $u = \alpha(\mathbf{1}_d + \eta)$ with $\alpha > 0$, $\eta \in V_d$ and $|\eta|_\infty \leq K/(Q - K)$. In particular, if $Q > 2K$ all coordinates of u have the same sign.*

(ii) *If moreover $Q > 2K$, $Qb_j \in \mathbb{Z}^d$ for all j , and $u \in \mathbb{Z}^d$ is primitive, then*

$$\alpha \leq Q^{d-1} \Delta e^{K/Q} \left(1 + \frac{K^2}{(Q-K)^2}\right)^{1/2},$$

$$\max u \leq Q^{d-1} \Delta e^{K/Q} \left(1 + \frac{K^2}{(Q-K)^2}\right)^{1/2} \left(1 + \frac{K}{Q-K}\right) \leq Q^{d-1} \Delta \left(1 + \frac{10K}{Q}\right).$$

Proof. (i) Write $u = \alpha \mathbf{1}_d + \eta'$ with $\eta' \in V_d$. From $\langle u, \ell_j \rangle = 0$ and $\langle \mathbf{1}_d, b_j \rangle = 0$ we get $Q\langle \eta', b_j \rangle + \alpha + \eta'_{r_j} = 0$ for all j . If $\alpha = 0$ then $\eta' = \sum_j \langle \eta', b_j \rangle b_j^* = -Q^{-1} \sum_j \eta'_{r_j} b_j^*$, so $|\eta'|_\infty \leq (K/Q)|\eta'|_\infty$ and $u = 0$, a contradiction. So $\alpha \neq 0$; take $\alpha > 0$ and $\eta = \eta'/\alpha$, so that $\langle \eta, b_j \rangle = -(1 + \eta_{r_j})/Q$ and $\eta = -Q^{-1} \sum_j (1 + \eta_{r_j}) b_j^*$, whence $|\eta|_\infty \leq Q^{-1}(1 + |\eta|_\infty)K$, i.e. $|\eta|_\infty \leq K/(Q - K)$, which is < 1 when $Q > 2K$.

(ii) Since $L \subseteq R(u)$, $\text{covol}(L) \geq \text{covol}(R(u)) = |u|_2 = \alpha|\mathbf{1}_d + \eta|_2 \geq \alpha\sqrt{d}$. The orthogonal projection $\pi: u^\perp \rightarrow V_d$ is bijective, as $\langle u, \mathbf{1}_d \rangle \neq 0$, and it multiplies covolumes by $\cos \vartheta = |\langle u, \mathbf{1}_d \rangle|/(|u|_2 \sqrt{d}) = \sqrt{d}/|\mathbf{1}_d + \eta|_2 \geq (1 + |\eta|_\infty^2)^{-1/2}$. Now $\pi(\ell_j) = Qb_j + \pi(e_{r_j})$ and $\pi(e_{r_j}) = \sum_i \langle e_{r_j}, b_i^* \rangle b_i = \sum_i (b_i^*)_{r_j} b_i$, so $\pi(L)$ has basis matrix $B(QI + A)$ with $A_{ij} = (b_i^*)_{r_j}$, where B is the basis matrix of Λ . Hence

$$\begin{aligned} \text{covol}(\pi(L)) &= Q^{d-1} \text{covol}(\Lambda) |\det(I + A/Q)| \\ &\leq Q^{d-1} \sqrt{d} \Delta \prod_i \left(1 + \frac{|b_i^*|_1}{Q}\right) \leq Q^{d-1} \sqrt{d} \Delta e^{K/Q}. \end{aligned}$$

Here the first inequality is Hadamard's, using $\sum_j |A_{ij}| \leq |b_i^*|_1$ (the rows r_j are distinct), and the second follows from (3). Therefore $\alpha\sqrt{d} \leq \text{covol}(L) = \text{covol}(\pi(L))/\cos \vartheta \leq Q^{d-1} \sqrt{d} \Delta e^{K/Q} (1 + |\eta|_\infty^2)^{1/2}$. The last inequality of the lemma is elementary for $x = K/Q \leq \frac{1}{2}$: $e^x \leq 1 + 1.3x$, $(1 + x^2/(1 - x)^2)^{1/2} \leq 1 + x$, $1 + x/(1 - x) \leq 1 + 2x$, and $(1 + 1.3x)(1 + x)(1 + 2x) \leq 1 + 10x$. $\square$

Proposition 12 (the constant $\mathbf{K}_s$). *Let $K_1 = \sum_{i \leq b-2} |b_i^*|_1$ for the base basis $b_i = \frac{1}{2} g x^i$ of Λ_1 , and let $c_i = \langle b_i^*, v_1 \rangle / h_1$. For the basis of Λ_s in Proposition 9, which is $\{b_i \otimes m\} \cup \{v_1 \otimes b'_j\}$ with b'_j the basis of Λ_{s-1} and $d' = b^{s-1}$, a dual basis is*

$$(b_i \otimes m)^* = b_i^* \otimes m - c_i \mathbf{1}_b \otimes \left(m - \frac{1}{d'} \mathbf{1}_{d'}\right), \quad (v_1 \otimes b'_j)^* = \frac{1}{h_1} \mathbf{1}_b \otimes (b'_j)^*.$$

Consequently the constant (3) for this basis of Λ_s equals

$$K_s = d' \sum_i \left(|b_i^* - c_i(1 - \frac{1}{d'}) \mathbf{1}_b|_1 + |c_i| b^{\frac{d'-1}{d'}} \right) + \frac{b}{h_1} K_{s-1} \leq d' \left(1 + \frac{4b}{3} \right) K_1 + \frac{2b}{3} K_{s-1}.$$

Proof. Direct verification of $\langle \cdot, \cdot \rangle = \delta$, using $\langle \mathbf{1}_b, b_i \rangle = 0$, $\langle \mathbf{1}_b, v_1 \rangle = h_1$, $\langle m - \frac{1}{d'} \mathbf{1}_{d'}, b'_j \rangle = (b'_j)_m$, and that all vectors lie in $V_{bd'}$. The ℓ^1 norms are read off blockwise; the crude bound uses $|c_i| \leq |b_i^*|_1 |v_1|_\infty / h_1 = \frac{2}{3} |b_i^*|_1$. $\square$

Numerically $K_1(5) = 424/55 \approx 7.71$, $K_1(7) \approx 14.19$, $K_1(11) \approx 32.20$, $K_1(13) \approx 44.03$, and, for instance, $K_2(5) \approx 98.25$, $K_3(11) \approx 1.78 \cdot 10^4$, $K_4(13) \approx 5.52 \cdot 10^5$. We checked the recursion against a direct computation of the dual basis for $(b, s) \in \{(5, 2), (7, 2), (5, 3)\}$.

4 The tensor perturbation

Fix $b \geq 5$ odd with $3 \nmid b$, $s \geq 1$, $d = b^s$, and $R_s, x_1, \dots, x_s$ as in Proposition 9.

Definition 13 (the lattice $\mathbf{L}_s(\mathbf{Q}, \rho)$). Let $Q \in 2^s \mathbb{Z}_{>0}$ and $\rho \in \mathbb{Z}_{>0}$. For $1 \leq k \leq s$ put

$$\Phi_k = \Phi_k^{(Q, \rho)} := \frac{Q}{2^k} (1 - x_k) \prod_{j \leq k} (2 + x_j) + \rho \prod_{j < k} x_j^{b-1} \in R_s,$$

and let $L_s(Q, \rho)$ be the $\mathbb{Z}$ -span of the $d - 1$ vectors $\ell_{k,i,m} := x_k^i m \Phi_k$, where $1 \leq k \leq s$, $0 \leq i \leq b-2$, and m runs over the monomials in $x_{k+1}, \dots, x_s$. We write $L_s(Q) = L_s(Q, 1)$.

By Proposition 9, $\ell_{k,i,m} = Q \beta_{k,i,m} + \rho \mu_{k,i,m}$, where the $\beta_{k,i,m}$ form the basis of Λ_s and $\mu_{k,i,m} = x_1^{b-1} \cdots x_{k-1}^{b-1} x_k^i m$ are *distinct* monomials (the variable x_k is the first one with exponent $\leq b-2$), none of them equal to $\prod_j x_j^{b-1}$. Thus $L_s(Q)$ is exactly the lattice of Section 3 for Λ_s with the tensor basis and distinct rows, and $L_s(Q, \rho) = \rho \cdot \text{span}\{(Q/\rho) \beta_{k,i,m} + \mu_{k,i,m}\}$ is the same construction with the real scale Q/ρ .

Definition 14 (level data). Set $\theta_1 = Q/2$ and $\rho_1 = \rho$. For $1 \leq k \leq s$, provided $\theta_k, \rho_k > 0$, put

$$\begin{aligned} \varphi_k &= \theta_k g + \rho_k \in R_b, & N_k &= N(\varphi_k), & \psi_k &= \text{adj}(\overline{\varphi_k}) \quad (\text{so } \psi_k \overline{\varphi_k} = N_k), \\ \gamma_k &= \text{cont}(\psi_k), & \tilde{u}_k &= \varepsilon_k x^{b-1} \psi_k / \gamma_k \quad (\varepsilon_k \in \{\pm 1\} \text{ with } \langle \tilde{u}_k, \mathbf{1}_b \rangle > 0), \\ a_k &= \langle \tilde{u}_k, x^{b-1} \rangle = \varepsilon_k \psi_{k,0} / \gamma_k, & c_k &= \langle \tilde{u}_k, 2 + x \rangle = 2\tilde{u}_{k,0} + \tilde{u}_{k,1}, \\ \theta_{k+1} &= \frac{Q}{2^{k+1}} c_1 \cdots c_k, & \rho_{k+1} &= \rho a_1 \cdots a_k \quad (k < s), \end{aligned}$$

where $\tilde{u}_{k,i}$ and $\psi_{k,i}$ denote the coefficients of x^i ; thus $\tilde{u}_{k,i} = \varepsilon_k \psi_{k,i+1} / \gamma_k$ with indices mod b . (Since $2^s \mid Q$, the θ_k are integers for $k \leq s$; the recursion is not continued beyond $k = s$.)

Lemma 15 (non-degeneracy). *If $Q \geq 8K_s \rho$, then for every $k \leq s$: $\theta_k, \rho_k > 0$, $N_k \neq 0$, and $\tilde{u}_k$ has all coordinates positive (so $a_k, c_k > 0$). Moreover $2\theta_k / \rho_k \geq Q/\rho$.*

Proof. For $\theta, \rho' > 0$ the element $\theta g + \rho'$ is not a zero divisor of $R_b \otimes \mathbb{Q}$: a root $\zeta \in \mu_b$ would satisfy $g(\zeta) = -\rho'/\theta \in \mathbb{Q}_{<0}$, but $g(\zeta) = 2 - \zeta - \zeta^2$ is rational only for $\zeta \in \{1, -1, \omega, \bar{\omega}\}$ (ω a primitive cube root of unity), and $\mu_b \cap \{1, -1, \omega, \bar{\omega}\} = \{1\}$ because b is odd and $3 \nmid b$, while $g(1) = 0$. Now induct on k . Given $\theta_k, \rho_k > 0$ we have $N_k \neq 0$ and $\psi_k \neq 0$. The lattice $L^{(k)} := \text{span}\{x^i \varphi_k : i \leq b-2\} = \rho_k \text{span}\{(2\theta_k/\rho_k)b_i + e_i\}$ is of the type of Lemma 11 with $K = K_1$ and real scale $Q'_k := 2\theta_k/\rho_k$, and its orthogonal complement is spanned by $x^{b-1}\psi_k$ (Proposition 16 below). If $Q'_k \geq 8K_1$, Lemma 11(i) shows that $\tilde{u}_k$ has coordinates of one sign, all in $[\alpha(1 - \frac{1}{7}), \alpha(1 + \frac{1}{7})]$, whence $c_k/a_k \geq 3 \cdot \frac{6/7}{8/7} > 2$. Since $K_s \geq (2b/3)^{s-1}K_1 \geq K_1$ by Proposition 12, we have $Q'_1 = Q/\rho \geq 8K_1$, and inductively $2\theta_{k+1}/\rho_{k+1} = (Q/2^k\rho) \prod_{j \leq k} (c_j/a_j) \geq Q/\rho \geq 8K_1$. $\square$

Proposition 16 (one level). *Let $\varphi \in R_b$ with $N(\varphi) \neq 0$ and $L = \text{span}\{x^i \varphi : 0 \leq i \leq b-2\} \subset \mathbb{Z}^b$. Then the vector of signed maximal minors of L is $\pm x^{b-1} \text{adj}(\bar{\varphi})$; hence $L^\perp$ is spanned by $x^{b-1} \text{adj}(\bar{\varphi})$ and $[\text{sat } L : L] = \text{cont } \text{adj}(\bar{\varphi})$. Moreover*

$$\begin{aligned} \text{cont } \text{adj}(\bar{\varphi}) = 1 &\iff \text{rank}_{\mathbb{F}_p}(m_\varphi) \geq b-1 \text{ for all primes } p \\ &\iff \deg_{\mathbb{F}_p} \gcd(\varphi, x^b - 1) \leq 1 \text{ for all primes } p. \end{aligned}$$

Proof. The columns of m_φ are $x^j \varphi$, $0 \leq j \leq b-1$, so the minors of L (delete row i) are the cofactors $C_{i,b-1}$ of m_φ ; i.e. $w = \pm(\text{row } b-1 \text{ of } \text{adj } m_\varphi) = \pm \text{adj}(m_\varphi)^T e_{b-1} = \pm \text{adj}(m_{\bar{\varphi}}) e_{b-1} = \pm x^{b-1} \text{adj}(\bar{\varphi})$, using $m_\varphi^T = m_{\bar{\varphi}}$ and the fact that $\text{adj}(m_{\bar{\varphi}})$ is multiplication by $\text{adj}(\bar{\varphi})$. The content of w is the index. For the equivalences: $\text{adj}(M) \equiv 0 \pmod{p}$ iff all $(b-1)$ -minors of M vanish mod p iff $\text{rank}_{\mathbb{F}_p} M \leq b-2$; and $\text{rank}_{\mathbb{F}_p} m_\varphi = \text{rank}_{\mathbb{F}_p} m_{\bar{\varphi}} = \dim_{\mathbb{F}_p}(\varphi \cdot \mathbb{F}_p[x]/(x^b - 1)) = b - \deg_{\mathbb{F}_p} \gcd(\varphi, x^b - 1)$, since the image of multiplication by φ in the principal ideal ring $\mathbb{F}_p[x]/(x^b - 1)$ is the ideal generated by $\gcd(\varphi, x^b - 1)$. $\square$

Theorem 17 (normal vector). *Let $Q \geq 8K_s\rho$ and let the level data be as in Definition 14. Then $L_s(Q, \rho)$ has rank $d-1$ and $L_s(Q, \rho) \subseteq R(u)$, where*

$$u := \tilde{u}_1 \otimes \cdots \otimes \tilde{u}_s, \quad u(x_1, \dots, x_s) = \prod_{k=1}^s \tilde{u}_k(x_k),$$

is a primitive vector of $\mathbb{Z}^d$ with positive coordinates.

Proof. Rank $d-1$: Lemma 10 for the real scale $Q/\rho > K_s$. Primitivity and positivity of u : the coordinates of u are the products $\prod_k \tilde{u}_{k,i_k}$, positive by Lemma 15, and the content of a tensor product of integer vectors is the product of the contents (Gauss's lemma), so $\text{cont } u = 1$.

Orthogonality. Fix (k, i, m) . Since $\langle u, \ell_{k,i,m} \rangle = [x^0](u \overline{\Phi_k} x_k^{-i} \bar{m}) = [x_k^i m](u \overline{\Phi_k})$, it suffices to show that every monomial of $u \overline{\Phi_k}$ whose $x_1, \dots, x_{k-1}$ -exponents vanish has x_k -exponent $b-1$. Now

$$\begin{aligned} u \overline{\Phi_k} &= \frac{Q}{2^k} \prod_{j < k} \left[\tilde{u}_j(x_j)(2 + x_j^{-1}) \right] \cdot \tilde{u}_k(x_k) \bar{g}(x_k) \cdot \prod_{j > k} \tilde{u}_j(x_j) \\ &\quad + \rho \prod_{j < k} \left[\tilde{u}_j(x_j) x_j^{-(b-1)} \right] \cdot \tilde{u}_k(x_k) \cdot \prod_{j > k} \tilde{u}_j(x_j), \end{aligned}$$

a sum of two products of univariate factors. Extracting, for each $j < k$, the part with x_j -exponent 0 amounts to replacing the j -th factor by its constant coefficient: $[x_j^0]\tilde{u}_j(2 + x_j^{-1}) = 2\tilde{u}_{j,0} + \tilde{u}_{j,1} = c_j$ and $[x_j^0]\tilde{u}_j x_j^{-(b-1)} = [x_j^0]\tilde{u}_j x_j = \tilde{u}_{j,b-1} = a_j$. Hence the relevant part of $u\overline{\Phi_k}$ is

$$(\theta_k \bar{g}(x_k) + \rho_k) \tilde{u}_k(x_k) \prod_{j>k} \tilde{u}_j = \overline{\varphi_k} \tilde{u}_k \prod_{j>k} \tilde{u}_j = \pm \frac{N_k}{\gamma_k} x_k^{b-1} \prod_{j>k} \tilde{u}_j(x_j),$$

by $\overline{\varphi_k} \psi_k = N_k$. All its monomials have x_k -exponent $b-1$, as required. $\square$

Theorem 18 (index formula). *Under the hypotheses of Theorem 17,*

$$[R(u) : L_s(Q, \rho)] = \prod_{k=1}^s \gamma_k^{b^{s-k}}.$$

In particular $L_s(Q, \rho)$ is primitive if and only if $\gamma_1 = \dots = \gamma_s = 1$.

Proof. Induction on s . For $s = 1$, $L_1(Q, \rho) = \text{span}\{x^i \varphi_1\}$ and the index is γ_1 by Proposition 16. Let $s \geq 2$, $d' = b^{s-1}$, and view $\mathbb{Z}^d = \mathbb{Z}^b \otimes \mathbb{Z}^{d'}$ with x_1 in the first factor. The family $k = 1$ spans $L^{(1)} \otimes \mathbb{Z}^{d'}$ with $L^{(1)} = \text{span}\{x_1^i \varphi_1\}$, whose orthogonal complement in $\mathbb{R}^b$ is $\mathbb{R}\tilde{u}_1$. For $k \geq 2$ write $\Phi_k = (2 + x_1) \otimes A_k + x_1^{b-1} \otimes B_k$ with $A_k = \frac{Q}{2^k} (1 - x_k) \prod_{2 \leq j \leq k} (2 + x_j)$ and $B_k = \rho \prod_{2 \leq j < k} x_j^{b-1}$ in $R_{s-1}(x_2, \dots, x_s)$. Decompose $\mathbb{R}^b \otimes \mathbb{R}^{d'} = W \oplus W^\perp$ with $W = \tilde{u}_1^\perp \otimes \mathbb{R}^{d'}$ and $W^\perp = \mathbb{R}\tilde{u}_1 \otimes \mathbb{R}^{d'}$, and identify $W^\perp$ with $\mathbb{R}^{d'}$ via $\tilde{u}_1/|\tilde{u}_1|_2 \otimes y \mapsto y$. The projection of $\ell_{k,i,m}$ ($k \geq 2$) onto $W^\perp$ is

$$\frac{1}{|\tilde{u}_1|_2} (\langle \tilde{u}_1, 2 + x_1 \rangle A_k + \langle \tilde{u}_1, x_1^{b-1} \rangle B_k) x_k^i m = \frac{1}{|\tilde{u}_1|_2} (c_1 A_k + a_1 B_k) x_k^i m,$$

and $c_1 A_k + a_1 B_k = \frac{c_1 Q/2}{2^{k-1}} (1 - x_k) \prod_{2 \leq j \leq k} (2 + x_j) + a_1 \rho \prod_{2 \leq j < k} x_j^{b-1}$, so these projections form, up to the factor $1/|\tilde{u}_1|_2$, the basis of $L' := L_{s-1}(Q', \rho')$ with $Q' = c_1 Q/2$ and $\rho' = a_1 \rho$ in the variables $x_2, \dots, x_s$. The level data of L' are $(\theta_2, \rho_2), \dots, (\theta_s, \rho_s)$: indeed $Q'/2 = \theta_2$, $\rho' = \rho_2$, and the recursion is the same. By Lemma 15, $Q'/\rho' = (c_1/a_1)(Q/2\rho) \geq Q/\rho \geq 8K_s \geq 8K_{s-1}$, so the induction hypothesis applies to L' : its normal vector is $u' = \tilde{u}_2 \otimes \dots \otimes \tilde{u}_s$ and $[R(u') : L'] = \prod_{k \geq 2} \gamma_k^{b^{s-k}}$. Since L' has rank $d' - 1$, the projection onto $W^\perp$ is injective on $L_{\geq 2} := \text{span}\{\ell_{k,i,m} : k \geq 2\}$; hence $L_s(Q, \rho) \cap W = L^{(1)} \otimes \mathbb{Z}^{d'}$, the projection of $L_s(Q, \rho)$ onto $W^\perp$ is $L'/|\tilde{u}_1|_2$, and

$$\text{covol } L_s(Q, \rho) = \text{covol}(L^{(1)} \otimes \mathbb{Z}^{d'}) \cdot \text{covol}(L'/|\tilde{u}_1|_2) = \text{covol}(L^{(1)})^{d'} \cdot \frac{\text{covol } L'}{|\tilde{u}_1|_2^{d'-1}},$$

where $\text{covol}(L^{(1)} \otimes \mathbb{Z}^{d'}) = \text{covol}(L^{(1)})^{d'}$ because $L^{(1)} \otimes \mathbb{Z}^{d'}$ is the orthogonal direct sum of the d' copies $L^{(1)} \otimes e_m$. As $u = \tilde{u}_1 \otimes u'$ is primitive, $\text{covol } R(u) = |u|_2 = |\tilde{u}_1|_2 |u'|_2$, so

$$[R(u) : L_s(Q, \rho)] = \frac{\text{covol } L_s(Q, \rho)}{\text{covol } R(u)} = \left(\frac{\text{covol } L^{(1)}}{|\tilde{u}_1|_2} \right)^{d'} \frac{\text{covol } L'}{|u'|_2} = \gamma_1^{d'} \prod_{k \geq 2} \gamma_k^{b^{s-k}}. \quad \square$$

Proposition 19 (exceptional primes). *Let $n_b = \prod_{0 < i < j < b} \text{Res}_z(\frac{z^b-1}{z-1}, 1+z^i+z^j)$ and let P_b be the set of odd primes dividing $b(2^b+1)n_b$. Then $n_b \neq 0$, so P_b is finite, and for $\varphi = \theta g + \rho$ with $\theta, \rho \in \mathbb{Z}$ and a prime p :*

- (a) *if $p \mid \gcd(\theta, \rho)$ then $\deg \gcd_{\mathbb{F}_p}(\varphi, x^b - 1) = b$;*
- (b) *if $p \mid \theta$ and $p \nmid \rho$ then $\deg \gcd_{\mathbb{F}_p}(\varphi, x^b - 1) = 0$;*
- (c) *if $p = 2$ and ρ is odd then $\deg \gcd_{\mathbb{F}_2}(\varphi, x^b - 1) = 0$;*
- (d) *if p is odd, $p \notin P_b$ and $p \nmid \gcd(\theta, \rho)$, then $\deg \gcd_{\mathbb{F}_p}(\varphi, x^b - 1) \leq 1$.*

Proof. For roots of unity, $1 + \zeta_1 + \zeta_2 = 0$ forces $\{\zeta_1, \zeta_2\} = \{\omega, \bar{\omega}\}$, which is impossible in μ_b because $3 \nmid b$; hence every factor of n_b is a nonzero integer. (a) and (b) are clear: $\varphi \equiv 0$, respectively a nonzero constant, mod p . (c): if θ is even, $\varphi \equiv \rho \equiv 1$; if θ is odd, $\varphi \equiv g + 1 \equiv 1 + x + x^2 \pmod{2}$, whose roots are the primitive cube roots of unity, which are not b -th roots of unity because $3 \nmid b$. (d): Let $\mathbb{F} = \overline{\mathbb{F}}_p$. Since $p \nmid b$, $x^b - 1$ is separable over $\mathbb{F}$; fix a prime $\mathfrak{p} \mid p$ of $\mathbb{Z}[\zeta_b]$; reduction modulo $\mathfrak{p}$ maps μ_b bijectively onto the roots of $x^b - 1$ in $\mathbb{F}$. As $x^b - 1$ is squarefree over $\mathbb{F}$, $\deg \gcd_{\mathbb{F}_p}(\varphi, x^b - 1)$ is the number of distinct common roots, so suppose $\zeta_1 \neq \zeta_2$ are common roots. If $p \mid \theta$ then $p \nmid \rho$ and $\varphi \equiv \rho$ has no roots. So $p \nmid \theta$ and $g(\zeta_1) = g(\zeta_2) = -\rho/\theta$; since $g(a) - g(c) = -(a - c)(1 + a + c)$, we get $1 + \zeta_1 + \zeta_2 = 0$ in $\mathbb{F}$. If $\zeta_1, \zeta_2 \neq 1$, lifting to $\zeta_i \in \mu_b$ gives $\mathfrak{p} \mid (1 + \zeta_1 + \zeta_2)$; writing $\zeta_1 = \eta^i$, $\zeta_2 = \eta^j$ for a primitive $\eta \in \mu_b$, the algebraic integer $1 + \eta^i + \eta^j$ divides $\text{Res}_z(\frac{z^b-1}{z-1}, 1+z^i+z^j)$, so $p \mid n_b$, a contradiction. If $\zeta_1 = 1$, then $\zeta_2 = -2$, so $(-2)^b = 1$ in $\mathbb{F}$, i.e. $p \mid 2^b + 1$, again a contradiction. $\square$

Theorem 20 (first level). *Let $b \geq 5$ be odd, $3 \nmid b$, and let $q \geq 4K_1$ be an integer with $\text{rad}(P_b) \mid q$. Then $L_1(2q)$ is primitive. Its normal vector $\tilde{u}_1 = \pm x^{b-1} \text{adj}(\overline{\varphi}_1)$, $\varphi_1 = qg + 1$, has positive coordinates forming a $(2q - K_1)$ -dissociated set with $\max \tilde{u}_1 \leq (2q)^{b-1} \Delta_{b,1}(1 + 5K_1/q)$, and for $2^l \leq 2q - K_1$ the lift $\{2^j \tilde{u}_{1,i}\}$ is a dissociated set of size lb with $f(lb) \leq \max \tilde{u}_1 / 2^{l(b-1)}$.*

Proof. By Proposition 16 it suffices to check $\deg \gcd_{\mathbb{F}_p}(\varphi_1, x^b - 1) \leq 1$ for all p . For odd $p \mid q$ use (b); for odd $p \nmid q$ we have $p \notin P_b$ and $\gcd(q, 1) = 1$, so (d) applies; $p = 2$ is (c). The remaining assertions are Lemmas 10 and 11 with $K = K_1$, $Q = 2q$, together with (2). $\square$

Example 21. Let $b = 5$ and $Q = 16$, so $\varphi_1 = 8g + 1 = 17 - 8x - 8x^2$ and $\overline{\varphi}_1 = 17 - 8x^3 - 8x^4$. One finds

$$\psi_1 = \text{adj}(\overline{\varphi}_1) = 62017 + 48704x + 49792x^2 + 53704x^3 + 52104x^4,$$

$N_1 = 266321 = 11^2 \cdot 31 \cdot 71$, with $\psi_1 \overline{\varphi}_1 = 266321$ in R_5 , $\gamma_1 = 1$, and $\tilde{u}_1 = x^4 \psi_1 = (48704, 49792, 53704, 52104, 62017)$. Here $K_1 = 424/55$, so Lemma 10 guarantees that the coordinates are 8-dissociated (they are in fact 16-dissociated but not 17-dissociated, by brute force), and $\max \tilde{u}_1 / Q^4 = 62017/65536 \approx 0.946$, compared with $\Delta_{5,1} = 11/16$; the ratio tends to $\Delta_{5,1}$ as $Q \rightarrow \infty$, e.g. it is 0.6906 for $Q = 8200$. The lift with $l = 3$ is the dissociated 15-element set $\{\tilde{u}_{1,i}, 2\tilde{u}_{1,i}, 4\tilde{u}_{1,i}\}$. Finally $a_1 = 62017$ and $c_1 = 2 \cdot 48704 + 49792 = 147200$ are coprime, so the second level can be built on top of this one.

5 Explicit dissociated sets

Theorem 22. *Let $b \geq 5$ be odd with $3 \nmid b$, $s \geq 1$, $d = b^s$, $Q \in 2^s\mathbb{Z}$ with $Q \geq 8K_s$, and suppose that the level data of Definition 14 (with $\rho = 1$) satisfy $\gamma_1 = \dots = \gamma_s = 1$. Then $u = \tilde{u}_1 \otimes \dots \otimes \tilde{u}_s$ is a vector of d positive integers whose coordinates are $(Q - K_s)$ -dissociated, and*

$$\max u = \prod_{k=1}^s \max_i \tilde{u}_{k,i} \leq Q^{d-1} \Delta_{b,s} \left(1 + \frac{10K_s}{Q}\right).$$

For every $l \geq 1$ with $2^l \leq Q - K_s$, the set $A = \{2^j u_i : 0 \leq j < l, 1 \leq i \leq d\}$ is dissociated, $|A| = ld$, and

$$f(ld) \leq \frac{\max u}{2^{l(d-1)}} \leq \Delta_{b,s} \left(1 + \frac{10K_s}{Q}\right) \left(\frac{Q}{2^l}\right)^{d-1}.$$

Proof. Theorems 17 and 18 give $L_s(Q) = R(u)$; Lemma 10 with $K = K_s$ (Proposition 12) gives $R(u) \cap (-Q - K_s, Q - K_s)^d = \{0\}$; Lemma 11(ii) bounds $\max u$; the product formula holds because all $\tilde{u}_{k,i} > 0$; finally apply (2). $\square$

The sizes ld in Theorem 22 form a sparse set. The following classical observation, used by Conway and Guy and by Bohman [5] to pass from one size to all larger sizes, fills the gaps at no cost.

Lemma 23 (monotonicity). *If A is dissociated with $|A| = n \geq 1$, then $2A \cup \{1\}$ is dissociated with $n + 1$ elements and maximum $2 \max A$. Consequently $f(n + 1) \leq f(n)$ for all $n \geq 1$, and for every $m \geq n$ the explicit set $A_m = 2^{m-n}A \cup \{2^j : 0 \leq j < m - n\}$ is dissociated with $|A_m| = m$ and $\max A_m / 2^m = \max A / 2^n$ (the added set is empty when $m = n$).*

Proof. The elements of $2A$ are even and 1 is odd, so $2A \cup \{1\}$ has $n + 1$ elements. If two subsets of $2A \cup \{1\}$ have the same sum, the two sums have the same parity, so either both subsets contain 1 or neither does; removing 1 and halving gives two subsets of A with the same sum, hence the same subsets. The maximum is $2 \max A$ because $\max A \geq 1$, and $2 \max A / 2^n = \max A / 2^{n-1}$; taking the infimum over A gives $f(n + 1) \leq f(n)$, and iterating the construction $m - n$ times gives A_m . $\square$

Corollary 24. *Table 1 and Lemma 23 give explicit dissociated sets A of every size n with*

$$\begin{aligned} \max A &< 0.158523 \cdot 2^n && \text{for every } n \geq 43\,923, \\ \max A &< 0.132535 \cdot 2^n && \text{for every } n \geq 1\,199\,562, \\ \max A &< 0.103037 \cdot 2^n && \text{for every } n \geq 3\,841\,966. \end{aligned}$$

In particular $f(n) \leq 0.206073$ for all $n \geq 3\,841\,966$.

Proof. The rows $(b, s) = (11, 3)$, $(13, 4)$ and $(17, 4)$ of Table 1 provide dissociated sets of sizes 43 923, 1 199 562 and 3 841 966 with $\max A / 2^{|A|}$ below 0.158523, 0.132535 and 0.103037 respectively (Theorem 22 with the certificates of Section 6); apply Lemma 23. $\square$

5.1 An arithmetic progression of good scalings

Write $Q = 2^{s+1}t$, where $t \in \mathbb{Z}_{>0}$. Run the recursion of Definition 14 formally, without dividing by contents and without sign normalisation:

$$\begin{aligned}\hat{\theta}_1 &= 2^s t, & \hat{\rho}_1 &= 1, & \hat{\varphi}_k &= \hat{\theta}_k g + \hat{\rho}_k, & \hat{\psi}_k &= \text{adj}(\overline{\hat{\varphi}_k}), \\ \hat{a}_k &= [x^0] \hat{\psi}_k, & \hat{c}_k &= 2[x^1] \hat{\psi}_k + [x^2] \hat{\psi}_k, \\ \hat{\theta}_{k+1} &= 2^{s-k} t \prod_{j \leq k} \hat{c}_j, & \hat{\rho}_{k+1} &= \prod_{j \leq k} \hat{a}_j & (1 \leq k < s).\end{aligned}$$

All these quantities lie in $\mathbb{Z}[t]$. Equivalently,

$$\hat{\theta}_{k+1} = \frac{1}{2} \hat{\theta}_k \hat{c}_k, \quad \hat{\rho}_{k+1} = \hat{\rho}_k \hat{a}_k. \quad (5)$$

The product formula above shows integrality despite the factor $1/2$. If the first $k - 1$ actual levels are primitive, then $(\theta_k, \rho_k) = \sigma_k(\hat{\theta}_k(t), \hat{\rho}_k(t))$ for a common sign $\sigma_k \in \{1, -1\}$. Indeed the signs chosen in Definition 14 enter both products in the same way, and $\text{adj}(-\varphi) = (-1)^{b-1} \text{adj}(\varphi) = \text{adj}(\varphi)$ since b is odd. If level k is primitive as well, then $(a_k, c_k) = \varepsilon_k(\hat{a}_k(t), \hat{c}_k(t))$ for another common sign.

Definition 25 (the coprimality condition $\mathbf{H}(\mathbf{b}, \mathbf{s})$). We say that $H(b, s)$ holds if, for every $1 \leq k < s$, the polynomials $\hat{a}_k, \hat{c}_k$ are coprime in $\mathbb{Q}[t]$; equivalently, $R_k := \text{Res}_t(\hat{a}_k, \hat{c}_k) \neq 0$ for every $1 \leq k < s$. For $s = 1$ the condition is empty.

The condition can in fact be checked at the first level alone. To see this, define homogeneous polynomials $\mathcal{A}_b, \mathcal{C}_b \in \mathbb{Z}[\theta, \rho]$ by

$$\begin{aligned}\mathcal{A}_b(\theta, \rho) &= [x^0] \text{adj}(\theta \bar{g} + \rho), \\ \mathcal{C}_b(\theta, \rho) &= 2[x^1] \text{adj}(\theta \bar{g} + \rho) + [x^2] \text{adj}(\theta \bar{g} + \rho).\end{aligned}$$

Both have degree $b - 1$. Put $\kappa_b = (2^b + 1)/3 \in \mathbb{Z}$. Since multiplication by $\bar{g}$ has kernel $\mathbb{R}\mathbf{1}_b$, its adjugate matrix is $\kappa_b \mathbf{1}_b \mathbf{1}_b^T$: the product of its nonzero eigenvalues is $b(2^b + 1)/3$. Equivalently, $\text{adj}(\bar{g}) = \kappa_b \sum_{i=0}^{b-1} x^i$ in R_b . Together with the adjugate of a scalar matrix, this gives the polynomial identities

$$\begin{aligned}\mathcal{A}_b(0, \rho) &= \rho^{b-1}, & \mathcal{C}_b(0, \rho) &= 0, \\ \mathcal{A}_b(\theta, 0) &= \kappa_b \theta^{b-1}, & \mathcal{C}_b(\theta, 0) &= (2^b + 1) \theta^{b-1}.\end{aligned} \quad (6)$$

These identities hold over $\mathbb{Z}$, so they remain valid after reduction modulo a prime. Let

$$r_b := \text{Res}_q(\mathcal{A}_b(q, 1), \mathcal{C}_b(q, 1)) \in \mathbb{Z}. \quad (7)$$

Theorem 26 (nonvanishing of the base resultant). *For every odd $b \geq 5$ with $3 \nmid b$, the polynomials $\mathcal{A}_b(q, 1)$ and $\mathcal{C}_b(q, 1)$ have no common complex root. Consequently $r_b \neq 0$ for every such b .*

Proof. Put $\varphi = 1 + qg$ and write

$$\psi = \text{adj}(\bar{\varphi}) = \sum_{i=0}^{b-1} \psi_i x^i.$$

At $q = 0$ we have $\psi = 1$, so $\mathcal{A}_b(0, 1) = 1$. Suppose henceforth that $q \neq 0$ and that

$$\psi_0 = 0, \quad 2\psi_1 + \psi_2 = 0. \quad (8)$$

First, $\psi \neq 0$, including when $N(\varphi) = 0$. Indeed, the values $g(\zeta)$ for $\zeta \in \mu_b$ are distinct. If $\zeta \neq \eta$ and $g(\zeta) = g(\eta)$, then

$$0 = -(\zeta - \eta)(1 + \zeta + \eta),$$

so $\zeta + \eta = -1$. Two unit complex numbers with this sum are the primitive cube roots of unity, contrary to $3 \nmid b$. The eigenvalues of multiplication by $\bar{\varphi}$ are $1 + qg(\zeta^{-1})$, $\zeta \in \mu_b$; thus at most one is zero. This matrix has rank at least $b-1$, and its adjugate is a nonzero circulant. Its first column, the coefficient vector of ψ , is therefore nonzero.

Set $\lambda = (1+2q)/q$, and extend the coefficients ψ_i periodically with period b . Extracting coefficients of x^i in $\bar{\varphi}\psi = N(\varphi)$ gives

$$\psi_{i+2} = \lambda\psi_i - \psi_{i+1} \quad (1 \leq i \leq b-1). \quad (9)$$

If $\psi_1 = 0$, then (8) gives $\psi_2 = 0$, and the recurrence forces every coefficient to vanish, a contradiction. We may therefore put $y_i = \psi_i/\psi_1$. Then

$$y_1 = 1, \quad y_2 = -2, \quad y_b = 0, \quad y_{b+1} = 1.$$

Let r, w be the roots of $z^2 + z - \lambda = 0$, so $r + w = -1$. If $r = w = -1/2$, the unique sequence satisfying (9) and the given first two values is

$$y_i = (3i - 2)(-1/2)^{i-1} \quad (1 \leq i \leq b+1).$$

In particular $y_b \neq 0$, a contradiction. If $r \neq w$, the same recurrence gives

$$y_i = \frac{(r-1)r^{i-1} - (w-1)w^{i-1}}{r-w} \quad (1 \leq i \leq b+1).$$

The equalities $y_b = 0$ and $y_{b+1} = 1$ imply, respectively, that $(r-1)r^{b-1} = (w-1)w^{b-1}$ and that their common value is 1. Hence

$$r^{b-1}(r-1) = 1, \quad w^{b-1}(w-1) = 1, \quad r + w = -1. \quad (10)$$

These three identities are impossible. To see this, r is a nonzero algebraic integer, being a root of the monic polynomial $z^b - z^{b-1} - 1$, and $w = -1 - r$ belongs to $\mathbb{Q}(r)$. For any embedding $\sigma : \mathbb{Q}(r) \hookrightarrow \mathbb{C}$, write $z = \sigma(r)$ and $v = \sigma(w) = -1 - z$. The identities (10) give

$$|z|^{b-1}|z-1| = |v|^{b-1}|v-1| = 1.$$

Writing $a = \operatorname{Re} z$, we also have

$$|v|^2 - |z|^2 = 2a + 1, \quad |v - 1|^2 - |z - 1|^2 = 3(2a + 1).$$

If $a > -1/2$, both factors in the product for v are strictly larger than the corresponding factors for z , which is impossible; if $a < -1/2$, the reverse inequalities give the same contradiction. Thus $a = -1/2$. In particular $|z - 1| \geq 3/2$, so

$$|\sigma(r)|^{b-1} \leq \frac{2}{3} < 1$$

for every embedding σ . The norm of r is a nonzero integer, but the product of the absolute values of all its conjugates is strictly less than 1, a final contradiction. Therefore (8) cannot hold, proving the theorem. $\square$

Proposition 27 (coprimality at every depth). *For every odd $b \geq 5$ with $3 \nmid b$, the condition $H(b, s)$ holds for every $s \geq 1$.*

Proof. By Theorem 26, $r_b \neq 0$. The only common zero of $\mathcal{A}_b$ and $\mathcal{C}_b$ in $\mathbb{C}^2$ is $(0, 0)$. Indeed, if $\rho \neq 0$, homogeneity would give a common zero $q = \theta/\rho$ of the two polynomials in (7); if $\rho = 0$ and $\theta \neq 0$, then $\mathcal{A}_b(\theta, 0) = \kappa_b \theta^{b-1} \neq 0$.

Consider the map

$$F_b(\theta, \rho) = \left(\frac{1}{2} \theta \mathcal{C}_b(\theta, \rho), \rho \mathcal{A}_b(\theta, \rho) \right). \quad (11)$$

It sends every nonzero pair in $\mathbb{C}^2$ to a nonzero pair. When $\theta = 0$ and $\rho \neq 0$, its second coordinate is $\rho^b \neq 0$; when $\rho = 0$ and $\theta \neq 0$, its first coordinate is $\frac{1}{2}(2^b + 1)\theta^b \neq 0$. When both coordinates of the input are nonzero, a zero output would require $\mathcal{A}_b = \mathcal{C}_b = 0$, which has just been excluded.

For any $z \in \mathbb{C}$, the initial pair $(2^s z, 1)$ is nonzero, and (5) iterates F_b . Thus, at every level,

$$(\hat{\theta}_k(z), \hat{\rho}_k(z)) \neq (0, 0), \quad (\hat{a}_k(z), \hat{c}_k(z)) \neq (0, 0).$$

The polynomials $\hat{a}_k, \hat{c}_k$ have no common complex root and are therefore coprime in $\mathbb{Q}[t]$. This proves $H(b, s)$ for every s . $\square$

We next make the finite set of congruence obstructions independent of s . As before, $\operatorname{rad}(P_b)$ denotes the product of the primes in the finite set P_b . By Theorem 26, the positive integer

$$M_b := \operatorname{rad}(P_b) |r_b| \quad (12)$$

is defined for every allowed b .

Lemma 28 (local avoidance). *Let p be an odd prime such that $p \notin P_b$ and $p \nmid r_b$. Then $\mathcal{A}_b, \mathcal{C}_b$ have no common nonzero zero in $\mathbb{F}_p^2$, and the map F_b of (11), reduced modulo p , sends every nonzero pair to a nonzero pair.*

Proof. The Sylvester adjugate identity gives polynomials $U, V \in \mathbb{Z}[q]$ with

$$U(q)\mathcal{A}_b(q, 1) + V(q)\mathcal{C}_b(q, 1) = r_b.$$

After reduction modulo p , the right side is nonzero. Thus the two univariate polynomials have no common root, even if their degrees decrease after reduction. Homogeneity excludes a common zero with $\rho \neq 0$. For $\rho = 0$ and $\theta \neq 0$, the value $\mathcal{A}_b(\theta, 0) = \kappa_b \theta^{b-1}$ is nonzero: every prime divisor of κ_b divides $2^b + 1$ and is in P_b .

The three cases in the proof of Proposition 27 now apply over $\overline{\mathbb{F}}_p$. At $\theta = 0$, the second coordinate of F_b is ρ^b ; at $\rho = 0$, the first is $\frac{1}{2}(2^b + 1)\theta^b$. Both boundary values are nonzero when the input is nonzero, because $p \nmid 2(2^b + 1)$. With $\theta\rho \neq 0$, a zero output would require a common zero of $\mathcal{A}_b, \mathcal{C}_b$. $\square$

Theorem 29. *Let $b \geq 5$ be odd with $3 \nmid b$, let $s \geq 1$, and let M_b be as in (12). If $t \equiv 0 \pmod{M_b}$ and $Q = 2^{s+1}t \geq 8K_s$, then $\gamma_1 = \dots = \gamma_s = 1$; hence $L_s(Q)$ is primitive and Theorem 22 applies. The same modulus M_b works for every $s \geq 1$.*

Proof. For $s = 1$, write $Q = 4t = 2q$. Then $q = 2t \geq 4K_1$ and $\text{rad}(P_b) \mid q$, so Theorem 20 applies. Suppose henceforth that $s \geq 2$.

We first record properties of the formal recursion modulo each prime. If $p \mid 2t$, then $\hat{\theta}_j = 2^{s+1-j}t \prod_{i < j} \hat{c}_i$ is divisible by p for every $1 \leq j \leq s$; for $p = 2$ this uses $s + 1 - j \geq 1$. Starting with $\hat{\rho}_1 = 1$, (6) gives

$$\hat{a}_j \equiv \hat{\rho}_j^{b-1} \not\equiv 0 \pmod{p}, \quad \hat{\rho}_{j+1} = \hat{\rho}_j \hat{a}_j \not\equiv 0 \pmod{p}.$$

Thus $\hat{\rho}_j$ is nonzero modulo p at every level. This argument does not divide by 2 in characteristic 2.

If $p \nmid 2t$, then p is odd, $p \notin P_b$ and $p \nmid r_b$, because $M_b \mid t$. The initial pair $(2^s t, 1)$ is nonzero modulo p . Lemma 28 and (5) show that $(\hat{\theta}_j(t), \hat{\rho}_j(t))$ is nonzero modulo p for every j .

Now induct on the actual level k , assuming $\gamma_j = 1$ for $j < k$. The observation preceding Definition 25 gives $(\theta_k, \rho_k) = \pm(\hat{\theta}_k(t), \hat{\rho}_k(t))$ with a common sign. If $p \mid 2t$, then $p \mid \theta_k$ and $p \nmid \rho_k$, so Proposition 19(b) gives degree zero for the common gcd. For every remaining prime, $p \notin P_b$ and $p \nmid \gcd(\theta_k, \rho_k)$, so Proposition 19(d) gives degree at most one. The level data are nondegenerate by Lemma 15, since $Q \geq 8K_s$. Proposition 16 therefore gives $\gamma_k = 1$, completing the induction. $\square$

Corollary 30. *Let $b \geq 5$ be odd with $3 \nmid b$, and let $s \geq 1$. Then*

$$\limsup_{l \rightarrow \infty} f(lb^s) \leq \Delta_{b,s}.$$

Proof. Fix b, s and the modulus M_b of Theorem 29. Put $W = 2^{s+1}M_b$. For each sufficiently large l , choose

$$Q = W \left\lceil \frac{2^l + K_s}{W} \right\rceil, \quad 2^l + K_s \leq Q < 2^l + K_s + W.$$

Then $Q \geq 8K_s$ and Theorems 29 and 22 give

$$f(lb^s) \leq \Delta_{b,s}(1 + 10K_s/Q)(Q/2^l)^{b^s-1} \longrightarrow \Delta_{b,s}.$$

□

Corollary 31. *In particular,*

$$\limsup_{l \rightarrow \infty} f(1331l) \leq \Delta_{11,3} < 0.31618, \quad \limsup_{l \rightarrow \infty} f(2197l) \leq \Delta_{13,3} < 0.30299.$$

Since f is nonincreasing, $\lim_{n \rightarrow \infty} f(n) \leq \Delta_{13,3}$; hence $f(n) < 0.303$ for all sufficiently large n .

Proof. Apply Corollary 30 and Lemma 23. The displayed numerical bounds follow from (1). □

Corollary 31 requires only numerical evaluation of (1). The explicit sets of Table 1 give the sharper effective statement $f(n) \leq 0.206073$ for all $n \geq 3\,841\,966$ (Corollary 24).

We give quantitative bounds before deriving a rate. In what follows, $\log$ denotes the natural logarithm, while $\log_2$ denotes the logarithm to base 2.

Lemma 32 (height and perturbation bounds). *For every allowed b , the modulus in (12) satisfies*

$$\log M_b \leq 4b^3.$$

For every $s \geq 1$ and $d = b^s$, the constants of Proposition 12 satisfy

$$K_1 \leq \frac{2(b^2 - 1)}{3}, \quad K_s \leq 5K_1d \leq \frac{10}{3}b^2d.$$

Proof. For an integer polynomial P , let $\|P\|_{\text{coeff},1}$ be the sum of the absolute values of its coefficients. This norm is submultiplicative. Every entry in the matrix of multiplication by $q\bar{g} + 1$ has this norm at most 3. Expanding a $(b-1) \times (b-1)$ minor as a determinant gives

$$\|\mathcal{A}_b(q, 1)\|_{\text{coeff},1}, \quad \|\mathcal{C}_b(q, 1)\|_{\text{coeff},1} \leq S_b := 3(b-1)!3^{b-1}.$$

The extra factor 3 allows for the linear combination defining $\mathcal{C}_b$. Both polynomials have degree $b-1$, since their leading coefficients are κ_b and $2^b + 1$, respectively, by (6). Every row of their Sylvester matrix has Euclidean norm at most S_b . Hadamard's inequality therefore gives

$$\log |r_b| \leq 2(b-1) \log S_b \leq 2b^2 \log(3b). \quad (13)$$

Each factor in the definition of n_b is a product, over $b-1$ complex roots of unity, of numbers of absolute value at most 3. Since these resultants are nonzero integers by Proposition 19,

$$\log \text{rad}(P_b) \leq \log b + \log(2^b + 1) + \binom{b-1}{2}(b-1) \log 3.$$

Combining this with (13) gives

$$\log M_b \leq \log b + \log(2^b + 1) + \binom{b-1}{2}(b-1)\log 3 + 2b^2 \log(3b) \leq 4b^3$$

for $b \geq 5$. For the last inequality, one may use $\log b \leq b$, $\log(2^b + 1) \leq b$, $\log(3b) \leq b$ and $\log 3 < 2$.

For the bound on K_1 , let $D_i \in V_b$ be the dual vectors to $e_i - e_{i+1}$, $0 \leq i \leq b-2$. Directly,

$$\|D_i\|_1 = \frac{2(i+1)(b-i-1)}{b}, \quad \sum_{i=0}^{b-2} \|D_i\|_1 = \frac{b^2-1}{3}.$$

The base basis is the image of this chain basis under $M = I + P/2$, where P is the cyclic shift. This map and its inverse preserve V_b , so $b_i^* = M^{-T}D_i$. The convergent geometric series for M^{-T} gives $\|M^{-T}\|_{1 \rightarrow 1} \leq \sum_{j \geq 0} 2^{-j} = 2$, proving the estimate for K_1 . Finally, write $E_s = K_s/b^s$. The recurrence in Proposition 12 gives

$$E_s \leq \left(\frac{1}{b} + \frac{4}{3}\right) K_1 + \frac{2}{3} E_{s-1}, \quad E_1 = K_1/b.$$

Induction yields $E_s \leq (4 + 3/b)K_1 \leq 5K_1$, which proves the result. $\square$

Theorem 33 (effective decay rate). $f(n) \leq n^{-c/\log \log n}$ for infinitely many n , with $c = 1/20$.

Proof. For each sufficiently large integer s , let $b = b(s)$ be the least odd integer not divisible by 3 which is at least $2s \log_2 s$. The gaps between consecutive positive integers congruent to 1 or 5 modulo 6 are at most 4, so

$$2s \log_2 s \leq b \leq 2s \log_2 s + 4 \leq s^2$$

for all sufficiently large s . Consequently $2^{-b}b^{s-1} \leq s^{-2s}s^{2s-2} = s^{-2} \leq 1/2$. With $d = b^s$, this implies

$$\log(\Delta_{b,s}(3/2)^s) = \frac{b^s - 1}{b - 1} \log(1 + 2^{-b}) \leq \frac{b}{b-1} 2^{-b}b^{s-1} < 1,$$

and hence $\Delta_{b,s} \leq e(2/3)^s$.

Put $W = 2^{s+1}M_b$ and choose

$$l = \lceil \log_2(W + 8K_s) \rceil + \lceil \log_2(200d) \rceil, \quad Q = W \left\lceil \frac{2^l + K_s}{W} \right\rceil. \quad (14)$$

Thus $2^l \geq 200d(W + 8K_s)$ and $2^l + K_s \leq Q < 2^l + K_s + W$. In particular, $Q \geq 8K_s$ and $Q/2^{s+1}$ is a multiple of M_b . Theorem 29 certifies all levels as primitive. The complete loss, including rounding to the progression, satisfies

$$\begin{aligned} \log((Q/2^l)^{d-1}(1 + 10K_s/Q)) &\leq \frac{(d-1)(K_s + W) + 10K_s}{2^l} \\ &\leq \frac{d(W + 8K_s)}{2^l} \leq \frac{1}{200}. \end{aligned}$$

Here the middle inequality holds for $d \geq 2$. Theorem 22 therefore produces a set of size $n_s = ld$ with

$$f(n_s) \leq e^{1+1/200}(2/3)^s < 3(2/3)^s. \quad (15)$$

Lemma 32 gives $\log_2 W \leq s+1+4b^3/\log 2$ and $\log_2 K_s \leq \log_2(10/3)+2\log_2 b+\log_2 d$. It follows directly from (14) that

$$l = O(b^3 + s + \log d).$$

Since $b \leq s^2$ and $\log d = s \log b \leq 2s \log s$, this upper bound is a fixed polynomial in s , whereas $d \geq 5^s$. In particular, for all sufficiently large s ,

$$d \leq n_s = ld \leq d^4. \quad (16)$$

All constants here are absolute, and the displayed estimates give effective upper bounds for the parameter choices.

As $\log b \leq 2 \log s \leq 2 \log \log d$, we have $s \geq \log d/(2 \log \log d)$. Combining (15) and (16) yields

$$f(n_s) \leq 3 \exp \left(-\frac{\log(3/2) \log d}{2 \log \log d} \right) \leq 3 n_s^{-\log(3/2)/(8 \log \log n_s)} \leq n_s^{-1/(20 \log \log n_s)}$$

for all sufficiently large s , because $\log(3/2)/8 > 1/20$. Finally, $n_s \geq 5^s$ tends to infinity, so these inequalities supply infinitely many distinct sizes. $\square$

Corollary 34. $f(n) \leq n^{-c'/\log \log n}$ for all sufficiently large n , with $c' = 1/50$.

Proof. For each sufficiently large s , use the size n_s produced above. Given large n , choose s maximal with $n_s \leq n$. Such an s exists and is finite, because $n_s \geq 5^s$; no monotonicity of the sequence (n_s) is needed. By maximality, $n < n_{s+1} \leq d_{s+1}^4$. Hence, for all sufficiently large s ,

$$\log n < 4(s+1) \log b(s+1) \leq 8(s+1) \log(s+1) \leq 16s \log s.$$

On the other hand, $n \geq n_s \geq 5^s$ implies $\log \log n \geq \log s + \log \log 5 > \log s$. Thus $s > \log n/(16 \log \log n)$. Lemma 23 and (15) give

$$f(n) \leq f(n_s) < 3(2/3)^s \leq 3 \exp \left(-\frac{\log(3/2) \log n}{16 \log \log n} \right) \leq n^{-1/(50 \log \log n)}$$

for all sufficiently large n , since $\log(3/2)/16 > 1/50$. $\square$

The progression in Theorem 29 supplies the specific gap bound W used in (14). This controls the distance from a good scaling to $2^l + K_s$, which is the quantity needed for the lifting estimate. The constants $1/20$ and $1/50$ retain substantial slack in the size estimates.

6 Computations and certificates

The algebraic certificates use exact integers and rational numbers in Python, with `sympy` for symbolic polynomial and matrix calculations. Floating-point quantities are used only for exploratory checks and displays explicitly marked as approximations. The table generator computes circulant adjugates by the Cayley–Hamilton identity; small cases are compared with fraction-free Bareiss elimination and direct symbolic adjugates.

Every one of the fifteen rows has a compressed exact witness. For each level the witness stores θ, ρ , the determinant, and the first two coefficients of $\psi = \text{adj}(\bar{\varphi})$, from which every other coefficient is exactly reconstructible. Writing $a = 2\theta + \rho$, the independent verifier works backwards through

$$a\psi_i = \theta(\psi_{i+1} + \psi_{i+2}) \quad (1 \leq i < b),$$

with subscripts modulo b , and checks every division and the cyclic closure. It independently computes

$$T_0 = 2, \quad T_1 = -\theta, \quad T_j = -\theta T_{j-1} + a\theta T_{j-2}, \quad N(\varphi) = a^b + T_b - \theta^b.$$

Indeed, if r_1, r_2 are the roots of $\theta z^2 + \theta z - a$, then $T_j = \theta^j(r_1^j + r_2^j)$, and taking the product over the b th roots of unity gives this determinant formula for odd b . Finally the verifier checks $a\psi_0 - \theta(\psi_1 + \psi_2) = N(\varphi) \neq 0$; thus it fixes the scale of the adjugate as well as its direction. It then reconstructs the tensor maximum, checks every content and the theorem’s inequalities, and verifies the decimal upper bounds. Full SHA-256 digests identify the witness files and the reconstructed integers. The data are in `results/certificates.json` and `results/witnesses/`; run `python tools/verify_certificates.py` from the repository root to check all rows independently of the production adjugate implementation. The repository is linked in the Acknowledgements.

1. *Correctness checks.* For $(b, s) \in \{(5, 1), (5, 2), (3, 2), (7, 2)\}$ and several Q we built the basis of $L_s(Q)$ explicitly, computed its vector of maximal minors directly, and verified that it equals $\pm \prod_k \gamma_k^{b^{s-k}} \cdot u$ with $u = \otimes_k \tilde{u}_k$, as predicted by Theorems 17 and 18 (for instance $(b, s) = (3, 2)$, $Q = 12$ gives $\gamma = (19, 169)$ and index $19^3 \cdot 169$; the case $3 \mid b$ is not covered by our theorems but the algebra of Section 4 still applies). The invariants h_s , $\text{covol}(\Gamma_s)$ and $\Delta_{b,s}$ of Proposition 9 were verified numerically. For $(b, s) = (3, 2)$, enumeration of $7^8 = 5\,764\,801$ coefficient vectors found no nonzero point violating (A1) in the chosen finite box, and no violation of (A2) at the three tested real parameters $t = 1, 5/4, 3/2$ in that box. These finite checks do not replace the proofs of Lemma 6 and Proposition 7, which apply to every real t .
2. *Brute-force dissociation.* The vectors of Definition 14, lifted with $b = 5$, $l \in \{3, 4, 5\}$ (for $Q = 16, 24, 40$) and with $b = 7$, $l = 3$ (for $Q = 24$), of sizes 15, 20, 25 and 21, as well as the 25-element vectors $u = \tilde{u}_1 \otimes \tilde{u}_2 \in \mathbb{Z}^{25}$ of Definition 14 for $(b, s) = (5, 2)$ and $Q \in \{108, 116, 124\}$ (scalings below the threshold $8K_2$ of Theorem 22, which therefore does not cover them), were verified to be dissociated by a meet-in-the-middle

enumeration of all signed sums. These are independent checks of small examples; the one-level examples also fail the sufficient divisibility or size hypotheses of Theorem 20, so that theorem is not being invoked to certify them.

3. *Sharpness of Lemma 10.* For $b = 5$ and $Q \in \{16, 24, 40\}$ the coordinates of $\tilde{u}_1$ are in fact exactly Q -dissociated, whereas the lemma guarantees $Q - K_1 \approx Q - 7.7$. We do not know whether $L_s(Q) \cap (-Q, Q)^d = \{0\}$ holds in general; if it did, one could take $Q = 2^l$ whenever 2^l is a good scaling, and the factor $(Q/2^l)^{d-1}$ in Theorem 22 would disappear.
4. *Level criterion.* For $b \in \{5, 7, 9, 11, 13\}$ and $8 \leq q < 40$ the criterion of Proposition 16, evaluated by factoring $\text{Res}(\varphi_1, x^b - 1)$ and computing polynomial gcds mod p , agreed with the directly computed index in all 160 cases. For $3 \mid b$ the first level is never primitive: $\varphi_1(\omega) = 3q + 1$ for both primitive cube roots of unity ω , so every prime factor of $3q + 1$ is exceptional; this is why we assume $3 \nmid b$.
5. *Theorem 20.* $P_5 = \{3, 5, 11\}$, $P_7 = \{3, 7, 43\}$, $P_{11} = \{3, 11, 23, 683\}$; for $q = m \cdot \text{rad}(P_b)$, $1 \leq m \leq 40$, the first level was primitive in all 120 cases. The level-one polynomials satisfy $\hat{a}_1(q) = N(q) - \frac{q}{b}N'(q)$ with $N(q) = \prod_{\zeta \neq 1} (1 + qg(\zeta))$, and $\text{Res}_q(\hat{a}_1, \hat{c}_1)$ equals $2 \cdot 11 \cdot 19$, $2^3 \cdot 7 \cdot 43 \cdot 379$, $2 \cdot 3^5 \cdot 5^2 \cdot 23^3 \cdot 683 \cdot 14797$ and $3^3 \cdot 13 \cdot 53^3 \cdot 61 \cdot 2731 \cdot 626592619$ for $b = 5, 7, 11, 13$; the primes 19 (for $b = 5$) and 7 (for $b = 7$) are exactly the second-level failures observed at $Q = 20$ and $Q = 12$.
6. *Checks of first-level coprimality.* The four nonzero resultants in the preceding item give numerical checks of Theorem 26, which proves nonvanishing for every allowed b without computation. The file `results/first_level_resultants.json` contains the complete integer coefficients of the first-level polynomials and adjugates, their determinants, and the resultant values and factorisations. The accompanying `exp09` script checks the polynomial adjugate identity and the resultants with exact arithmetic. The separate `exp10` script checks the circulant and recurrence identities and computes polynomial gcds for all odd $3 \leq b \leq 101$, including $3 \mid b$ as a control. These calculations supplement the general proof.
7. *The table.* For each (b, s) in Table 1 we chose $l = \lceil \log_2(200(d-1)K_s) \rceil$ as an initial scale, and the least t with $Q = 2^{s+1}t \geq 2^l + K_s$ and $\gamma_1 = \dots = \gamma_s = 1$; at most three values of t were needed in every case. Rounding and search are accounted for using the actual value of Q . Put $z = (d-1)(Q - 2^l)/2^l$. The verifier checks the rational inequalities $0 \leq z < 1$ and $(1-z)^{-1} < 1.006$ in every row. The binomial theorem gives

$$(Q/2^l)^{d-1} \leq \sum_{j \geq 0} z^j = (1-z)^{-1} < 1.006.$$

The exact maximum is also checked against the shape estimate in every row. For (17, 4) the four adjugates involve integers of about $3 \cdot 10^6$ bits. The two bound columns round the exact rationals $\max u/2^{l(d-1)}$ and $\max u/2^{l(d-1)+1}$ upwards to six decimal places.

For a positive rational p/q , the printed bound is $\lceil 10^6 p/q \rceil / 10^6$, computed entirely with integers. The $\Delta_{b,s}$ column is an approximation to five decimal places, not an asserted upper bound.

b	s	$d = b^s$	l	$n = ld$	Q	$f(n) \leq$	$\max A/2^n \leq$	$\Delta_{b,s}$
5	1	5	13	65	8200	0.690649	0.345325	0.68750
7	1	7	15	105	32784	0.674015	0.337008	0.67188
11	1	11	16	176	65572	0.670804	0.335402	0.66699
13	1	13	17	221	131120	0.669767	0.334884	0.66675
5	2	25	19	475	524392	0.537132	0.268566	0.53457
7	2	49	22	1078	4194584	0.474515	0.237258	0.47299
11	2	121	25	3025	33555528	0.448813	0.224407	0.44706
13	2	169	26	4394	67110656	0.447207	0.223604	0.44520
5	3	125	25	3125	33555168	0.771243	0.385622	0.76915
7	3	343	28	9604	268438368	0.463429	0.231715	0.46171
11	3	1331	33	43923	8589952400	0.317045	0.158523	0.31617
13	3	2197	34	74698	17179903936	0.304339	0.152170	0.30299
17	3	4913	37	181781	137439055184	0.298073	0.149037	0.29699
13	4	28561	42	1199562	4398047063424	0.265070	0.132535	0.26412
17	4	83521	46	3841966	70368746289408	0.206073	0.103037	0.20556

Table 1: Certified explicit dissociated sets A of size $n = ld$ (Theorem 22). The set A is the lift of $u = \otimes_k \tilde{u}_k$; u is determined by (b, s, Q) through Definition 14, and the certificate is $\gamma_1 = \dots = \gamma_s = 1$. The previous explicit record is $\max A \leq 0.22002 \cdot 2^n$ [5].

Table 1 is read as follows. In every row the column $f(n) \leq$ is within 0.6% of $\Delta_{b,s}$ (the largest ratio, about 1.006, occurs at $(11, 1)$), so in these examples the finite-scale loss relative to $\Delta_{b,s}$ is small. The analytic upper bound on the right of Theorem 22 is at least $\Delta_{b,s} > (2/3)^s$. Thus that upper bound, with $s \leq 2$, cannot establish $\max A/2^n < \frac{1}{2}(2/3)^2 = 2/9$, or improve Bohman's 0.22002; for $s = 3$ one needs $2^{-b}b^2$ small, and $(7, 3)$ still gives 0.2317 while $(11, 3)$ is the first pair below 0.22002. Increasing s further requires b to grow like $s \log_2 s$ to keep $(1+2^{-b})^{(b^s-1)/(b-1)}$ bounded, which is the mechanism behind Theorem 33; the price is the dimension $d = b^s$ of the vector and the height of the adjudgates, about $3 \cdot 10^6$ bits for $(17, 4)$. The certificate itself is always s adjudgates of $b \times b$ integer matrices, and each row was obtained with at most three trial values of t . No row is claimed to be optimal for its size; by Lemma 23, every row is also an upper bound for $f(n)$ at all larger n (Corollary 24).

7 Remarks and open problems

7.1 Sizes reached by the construction

For fixed (b, s) , the binary lift produces sizes $n = ld$, where $d = b^s$. The scale required for a small loss depends on both the perturbation constant K_s and the spacing of good scalings.

More precisely, suppose that a good scaling can be found in every interval $[X, X + G_{b,s}]$ above the threshold $8K_s$. Choosing such a scaling with

$$2^l + K_s \leq Q \leq 2^l + K_s + G_{b,s}$$

gives

$$\left(\frac{Q}{2^l}\right)^{d-1} \leq \exp\left(\frac{(d-1)(K_s + G_{b,s})}{2^l}\right).$$

Thus $d(K_s + G_{b,s})/2^l = o(1)$ suffices to make the finite-scale loss in Theorem 22 tend to 1. Positive density of good scalings in intervals $[X, 2X]$ alone gives no such bound on their gaps near the prescribed point $2^l + K_s$.

Lemma 32 proves $K_s = O(b^2 d)$ with an absolute constant. For example, a polynomial bound $G_{b,s} = d^{O(1)}$ permits $l = O(\log d)$ and hence $n = O(d \log d)$ with a vanishing finite-scale loss. Such a size estimate requires control of $G_{b,s}$ in addition to control of K_s .

Lemma 23 extends any constructed set to every larger size without changing its normalized maximum. It is useful to compare this operation with other elementary extensions using their exact ratios. Write

$$\mathcal{R}(A) := \frac{\max A}{2^{|A|-1}}$$

for an individual dissociated set A .

Adjoining an element larger than all subset sums. If $|A| = n$, $S = \sum_{a \in A} a$, and $N = \max A$, then $A^+ = A \cup \{S + 1\}$ is dissociated and

$$\frac{\mathcal{R}(A^+)}{\mathcal{R}(A)} = \frac{S + 1}{2N}.$$

This factor depends on A . It is about $n/2$ when $S \sim nN$, whereas it equals 1 for $A = \{2^j : 0 \leq j < n\}$. The identity concerns the two individual sets; it does not assert that the extremal function f is multiplied by this factor.

Mixed binary lifts. Let u have d positive coordinates and be T -dissociated, with $T = 2^{l+1}$. Give r coordinates $l + 1$ binary digits and the remaining $d - r$ coordinates l digits, where $1 \leq r \leq d$. The resulting set A is dissociated, has $ld + r$ elements, and satisfies

$$\mathcal{R}(A) \leq 2^{d-r} \frac{\max u}{T^{d-1}}.$$

Thus the shape estimate $\max u / T^{d-1} = \Delta(1 + o(1))$, when it is available at this scale, yields the bound $f(ld + r) \leq 2^{d-r} \Delta(1 + o(1))$. The factor is a bound arising from this choice of common scale; it need not be the actual ratio of the mixed lift to a particular binary lift.

Lifts using a general dissociated digit set. Let $B \subset \mathbb{N}$ be dissociated, with $m = |B|$, $S = \sum_{a \in B} a$, and $H = \max B$. Let u be a positive Q -dissociated vector of length d , where Q is an integer and $Q \geq S + 1$. Then

$$A = B \cdot U := \{au_i : a \in B, 1 \leq i \leq d\}$$

is dissociated and has md elements: the difference between two digit sums in each coordinate has absolute value at most $S < Q$, and dissociation of B handles equality within each coordinate. Its normalized maximum satisfies the exact identity

$$\mathcal{R}(A) = \frac{H}{2^{m-1}} \left(\frac{Q}{2^m} \right)^{d-1} \frac{\max u}{Q^{d-1}}. \quad (17)$$

The 2^m distinct subset sums of B lie among the $S + 1$ integers $0, \dots, S$, so $S + 1 \geq 2^m$. Equality holds precisely for $B = \{1, 2, 4, \dots, 2^{m-1}\}$: if every integer in $[0, S]$ is a subset sum, the increasing elements of B must successively be one more than the sum of their predecessors. For this binary digit set, choosing $Q = S + 1 = 2^m$ makes the first two factors in (17) equal to 1. For each fixed nonbinary B , the factor $(Q/2^m)^{d-1}$ grows exponentially with d at every permitted scale $Q \geq S + 1$. The additional factor $H/2^{m-1}$ must also be retained; there is no general loss formula depending only on $S/2^m$.

Padding the lattice. For an admissible pair (Λ, v) in $\mathbb{R}^d$ of positive height h , set

$$\Lambda^+ = \{(\lambda + kv, -kh) : \lambda \in \Lambda, k \in \mathbb{Z}\} \subset V_{d+1}, \quad v^+ = e_{d+1}.$$

This is an admissible pair. Indeed, if the first d coordinates of $(\lambda + kv, -kh) + tv^+$ lie in $(-1, 1)^d$, condition (A2) for the original pair forces $|k| < 1$, hence $k = 0$; condition (A1) then gives $\lambda = 0$. The last coordinate consequently gives $|t| < 1$. The projection of $(v, -h)$ onto the orthogonal complement of $V_d \times \{0\}$ has length $h\sqrt{(d+1)/d}$, so

$$\text{covol}(\Lambda^+) = \text{covol}(\Lambda) h \sqrt{\frac{d+1}{d}}, \quad h^+ = 1, \quad \Delta^+ = h\Delta.$$

This is an exact factor. It exceeds 1 for the pairs used here, whose heights are $(3/2)^s$; an arbitrary admissible pair is only assumed to have $h > 0$.

Monotonicity also shows that the limit of $f(n)$ exists. Every certified row of Table 1 bounds $f(n)$ at every larger size, while an asymptotic rate requires quantitative control of how the certified sizes grow.

7.2 The exponent and heights of admissible pairs

Proposition 35. *Let (Λ, v) be admissible in $\mathbb{R}^b$, of positive height h , and write $\text{covol}(\Gamma) = 1 + \eta$. Its s -fold tensor power is admissible in dimension $d = b^s$, with normalized covolume*

$$\Delta_s = (1 + \eta)^{(b^s - 1)/(b - 1)} h^{-s}.$$

Conversely, every admissible pair in $\mathbb{R}^b$ satisfies $\Delta \geq b^{-1/2}$, and therefore $h \leq (1 + \eta)\sqrt{b}$.

Proof. The tensor formula follows by iterating Proposition 7. For the second assertion, (A1) and Minkowski's theorem give

$$\text{covol}(\Lambda) \geq 2^{-(b-1)} \text{vol}_{b-1}((-1, 1)^b \cap V_b) \geq 1,$$

where the last inequality is Vaaler's cube slicing theorem [15]. Divide by $\sqrt{b}$ and use $h\Delta = \text{covol}(\Gamma) = 1 + \eta$. $\square$

For our base pair, $h = 3/2$ and $\eta = 2^{-b}$. At the level of lattice covolumes, the height contributes the factor h^{-s} ; thus $\log h$ governs this exponential decay in s . An improved height can give an improved bound for f , provided the corresponding primitivisation and lift have suitably controlled size and loss.

More precisely, suppose that a family of base pairs and tensor powers has $d = b^s \rightarrow \infty$,

$$h \geq b^\alpha, \quad \eta \frac{b^s - 1}{b - 1} \leq \log 2.$$

Then Proposition 35 gives $\Delta_s \leq 2d^{-\alpha}$. Suppose in addition that effective primitivisations and binary lifts produce sizes $n = ld$ satisfying

$$l = d^{o(1)}, \quad f(ld) \leq d^{o(1)} \Delta_s.$$

These two conditions imply

$$f(n) \leq d^{-\alpha+o(1)} = n^{-\alpha+o(1)}$$

along the constructed sizes. In the setting of Theorem 22, the second condition amounts to controlling its finite-scale factor $(Q/2^l)^{d-1}(1 + 10K_s/Q)$ by $d^{o(1)}$, in addition to satisfying $2^l \leq Q - K_s$. To obtain the same exponent for all sufficiently large integers, it is enough that the constructed sizes can be listed increasingly as $n_j \rightarrow \infty$ with $\log n_{j+1}/\log n_j \rightarrow 1$; Lemma 23 then interpolates between consecutive sizes. Monotonicity alone does not preserve this exponent across arbitrarily large multiplicative gaps. The height bound in Proposition 35 gives $\alpha \leq 1/2$ in this regime, in agreement with the lower bound $f(n) \gg n^{-1/2}$.

Problem 36. Determine

$$h^*(b) = \sup\{h : (\Lambda, v) \text{ is admissible in } \mathbb{R}^b, \text{covol}(\Gamma) \leq 1 + 2^{-b}\}.$$

Does $h^*(b) \rightarrow \infty$? Is $h^*(b) \geq b^\alpha$ for some $\alpha > 0$ and all sufficiently large b ?

There is a precise packing interpretation with an extension condition. Put

$$C_b = [-1/2, 1/2]^b \cap V_b, \quad \nu_b = \text{vol}_{b-1}(C_b).$$

Condition (A1) says that Λ is a packing lattice for C_b ; its packing density is

$$\delta(\Lambda) = \frac{\nu_b}{\text{covol}(\Lambda)} = \frac{\nu_b h}{\sqrt{b} \text{covol}(\Gamma)}.$$

Restrict to those packing lattices that admit a vector v satisfying the real- t condition (A2) and $1 \leq \text{covol}(\Lambda \oplus \mathbb{Z}v) \leq 1 + 2^{-b}$. Within this class, the exact identity

$$h = \frac{\sqrt{b} \text{covol}(\Gamma)}{\nu_b} \delta(\Lambda)$$

relates large heights to large packing densities, with the covolume factor between 1 and $1 + 2^{-b}$. In particular, $h \geq b^\alpha$ implies $\delta(\Lambda) \geq \nu_b b^{\alpha-1/2}/(1 + 2^{-b})$. A packing lattice satisfying (A1) alone need not supply either the extension vector or the required full-lattice covolume, so no equivalence with unrestricted lattice packing is asserted.

7.3 Siegel's lemma

As noted in [3], and observed earlier by Schinzel and Aliev [2], for $d \geq 2$ let C_d denote the optimal constant in Siegel's lemma for one equation in d unknowns (so that every nonzero $a \in \mathbb{Z}^d$ has a nonzero integer relation x with $|x|_\infty^{d-1} \leq C_d |a|_\infty$; see [6]). Then $f(d) \geq C_d^{-1}$. Each vector u of Theorem 22 is an explicit integer vector of height at most $Q^{d-1} \Delta_{b,s} (1 + 10K_s/Q)$ all of whose nonzero integer relations c satisfy $|c|_\infty \geq Q - K_s$, so it gives the explicit lower bound $C_d \geq (1 - K_s/Q)^{d-1} / (\Delta_{b,s} (1 + 10K_s/Q))$ for $d = b^s$; for example $C_{1331} \geq 3.15$ and $C_{83521} \geq 4.84$.

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